# Instruction Routing in Fine-Tuned Vision-Language-Action Models: A Cross-Architecture Analysis

**Anonymous Author(s)**
Affiliation
Address
email

**Abstract:** When vision-language models are fine-tuned for robotic action prediction, which representational changes are associated with training and which are inherited from the pretrained backbone? We compare trained and reference checkpoints of three Vision-Language-Action models across four LIBERO task suites via layer-wise probes, attention profiling, and counterfactual perturbations. We find that VLM-to-VLA fine-tuning is associated with a consistent exit-layer collapse in action-probe decodability ($\Delta R^2 = 0.16$–$0.46$) that is absent in VLA-to-VLA transitions ($\Delta R^2 < 0.10$) and in three NLP instruction-tuning controls ($\Delta R^2 = 0.000$), suggesting this reorganization is specific to action-oriented training rather than a generic fine-tuning artifact. Complementary analyses reveal that trained VLAs develop substantial sensitivity to instruction content while the dominant attention patterns at early layers largely pre-exist in pretrained backbones. These findings provide a cross-architecture decomposition of instruction routing in fine-tuned VLAs, disentangling training-associated representational changes from inherited structure.

**Keywords:** Vision-Language-Action models, representational analysis, attention routing, fine-tuning

## 1 Introduction

Vision-language-action models (VLAs) fine-tune pretrained language models on robot demonstrations to produce actions from visual observations and language instructions [1, 2], but which of the resulting representational changes reflect action training and which are inherited from the pretrained backbone? In these decoder-only architectures, image tokens precede instruction tokens, and the causal attention mask restricts multimodal integration to a single readout token [3, 4]; how representations at this critical position are reshaped by action fine-tuning remains unknown.

Behavioral studies show that VLAs exhibit visual dominance [5] and underweight instructions [6], but existing representational analyses target single architectures. No prior work tests whether instruction routing mechanisms are consistent across backbone families or decomposes training-associated effects from inherited base-model priors.

We address this gap with a cross-architecture analysis of three VLAs (OpenVLA [2], TraceVLA [7], and NORA [8]) across four LIBERO [9] task suites (§4.1). TraceVLA and NORA baselines are pure VLMs ($N = 2$ independent VLM-to-VLA transitions); OpenVLA's baseline is the OXE-pretrained [10] VLA (VLA-to-VLA transition).

Our analysis reveals three findings (Figure 1): (1) VLM-to-VLA fine-tuning is associated with substantial $R^2$ exit collapse ($\Delta R^2 = 0.16$–$0.46$), whereas VLA-to-VLA fine-tuning and reference models show only small fluctuations ($\Delta R^2 < 0.10$), with three NLP controls (all $\Delta R^2 = 0.000$) confirming transition-type specificity; (2) all trained VLAs exhibit medium-to-large counterfactual sensitiv-

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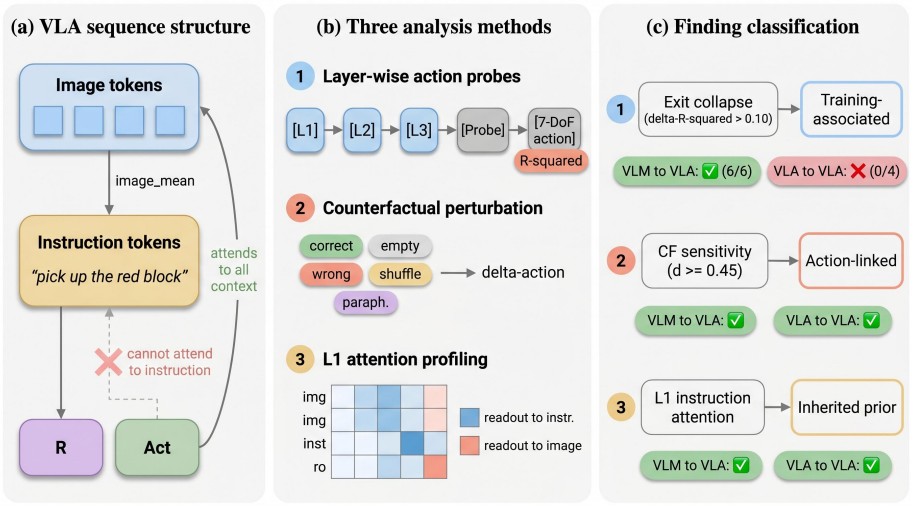



Figure 1: **Analysis framework.** We analyze hidden states at image-token averages and the readout token via layer-wise action probes, counterfactual instruction perturbations, and L1 attention profiling to distinguish training-associated changes from inherited backbone priors.

ity ($d = 0.45$–$1.52$); and (3) L1 instruction attention reflects base-model priors, not VLA training (§4.3–4.5). While concurrent work evaluates VLA instruction-following behaviorally [11, 12], we provide complementary representational analysis of how instruction information is routed internally. Our contributions are:

1. A cross-architecture representational analysis of three VLAs, combining trained/reference controls with token-level probing across a $3 \times 4$ architecture-suite matrix.
2. Evidence that $R^2$ exit collapse is associated with VLM-to-VLA transitions ($\Delta R^2 > 0.10$ across two architectures) but not with VLA-to-VLA fine-tuning or NLP instruction tuning of the same backbones.
3. A decomposition of VLA instruction routing into training-associated effects and inherited base-model priors, based on cross-architecture trained/reference contrasts.

## 2 Related Work

**Vision-language-action models.** VLAs fine-tune large language models on robot demonstrations to produce actions from visual and language input [1, 2, 13, 14, 7, 8, 15]. NLP instruction tuning is well-characterized behaviorally [16, 17], but the representational effects of VLA fine-tuning remain unexamined. Fine-tuning operates in a low intrinsic dimension [18] and can distort pretrained features [19]; recent VLA-specific work proposes representation-preserving strategies [20, 21]. Existing analyses target single architectures and cannot separate training dynamics from backbone artifacts.

**Mechanistic and behavioral analysis.** Mechanistic interpretability has revealed residual stream communication [22], induction heads [23], and task circuits [24, 25]; causal tracing localizes factual recall [26]; attention probing shows heads encode syntactic relations [27] and most can be pruned [28]. In VLMs, recent work characterizes visual token redundancy [29], progressive visual-textual alignment [30, 31], and instruction tokens as shallow-layer modality arbitration anchors [32]. Behavioral evaluations of VLAs reveal visual dominance [5, 33, 34]; mechanistic studies identify sparse semantic directions [6], interpretable features [12, 35], symbolic representations [36, 37], comprehension-action trade-offs [38], and task-specific attention heads for selective fine-tuning [39]. Concurrent continual-learning studies find that pretrained VLAs resist forgetting under sequential fine-tuning [40], consistent with our observation that VLA-to-VLA transitions preserve exit-layer structure. All prior representational analyses target a single architecture; our $3 \times 4$ cross-architecture matrix tests whether mechanisms replicate across backbone families.

**Concurrent work.** Fang et al. [11] construct a behavioral benchmark of counterfactual instruction failures on LIBERO; their evaluation measures *whether* the model follows the correct instruction, while our perturbation measures *how much* hidden-state geometry changes (§4.4). Buurmeijer et al. [35] probe low-level motor features in OpenVLA; instruction-level routing is the focus of our analysis. Concurrent work addresses visual representation collapse [41] and instruction information collapse [42] during VLA fine-tuning; our $\Delta R^2$ metric offers a complementary diagnostic for measuring such collapse across architectures. We additionally distinguish training-associated from inherited effects across three architectures, a decomposition absent from prior single-architecture studies.

**Probing and similarity methodology.** RSA [43], CKA [44], and linear probes [45, 46] provide complementary geometric and functional evaluation. We adopt RSA and MLP probes (CKA in appendix); the two metrics can diverge on the same data (Appendix A.5), and Cohen's $d$ vs. cosine distance are not comparable across trained/reference conditions (§4.4).

# 3 Method

We investigate how action-relevant representations are organized across three VLA architectures using complementary probing, attention, and perturbation analyses.

## 3.1 Architectures and Data

We study three VLAs that span distinct backbone families (Table 3). **OpenVLA** [2, 13] chains a frozen DinoSigLIP vision backbone [47, 48] (DINOv2 ViT-L/14 + SigLIP-SO400M), an MLP projector, and Llama-2-7B [49] (6.74B parameters, 32 layers, 32 multi-head attention [MHA] heads) for autoregressive 7-D action decoding. It is pretrained on Open X-Embodiment [10] (OXE; 970K episodes across multiple robot platforms). **TraceVLA** [7] pairs a SigLIP encoder with Phi-3-mini [50] (3.8B parameters, 32 layers, 32 MHA heads). TraceVLA is trained on an OXE subset with spatial trace conditioning. **NORA** [8] uses Qwen2.5-VL-3B ($\sim$3B parameters, 36 layers, grouped-query attention [GQA] with 16 query heads and 2 KV heads) with the native Qwen2.5-VL vision encoder, which processes images at dynamic resolution without a separate projection stage.

Each architecture is evaluated in trained and reference conditions; we define the *reference checkpoint* as the corresponding pretrained model before LIBERO fine-tuning. A critical asymmetry distinguishes the reference baselines: the OpenVLA reference is an OXE-pretrained VLA, already fine-tuned for action prediction on 970K episodes, while the TraceVLA and NORA references are pure vision-language models (Phi-3-Vision and Qwen2.5-VL) with no action training. OpenVLA trained-vs-reference comparisons therefore measure a VLA-to-VLA transition; TraceVLA and NORA comparisons measure VLM-to-VLA transitions. For OpenVLA, two additional conditions support deep-dive analyses (§4.2): the Llama-2 base language model and a random-weight initialization.

We evaluate on LIBERO [9], comprising four task suites (Goal, Spatial, Object, Long-Horizon) with 10 tasks and 50 demonstrations each. The three architectures and four suites form a $3 \times 4$ matrix; 10 of 12 cells are filled (NORA Object and Long-Horizon lack available checkpoints).

Architecture details appear in Table 3 (Appendix).

## 3.2 Representation Extraction

From each architecture, we extract hidden states at every transformer layer from two token positions: (a) **image_mean**, the mean activation over image tokens, and (b) **readout**, the hidden state at the final pre-action token, which follows both image and instruction tokens in the causal sequence and can attend to all preceding context. L0 denotes the embedding layer output, L1 through L$N$ the hidden states after each transformer layer ($N = 32$ for OpenVLA and TraceVLA, $N = 36$ for NORA), with the final layer including the output layer norm. In all three architectures, image tokens

precede instruction tokens; the causal mask prevents image tokens from attending to instructions by construction.

We cross two factors in a $2 \times 2$ factorial design: *training status* (trained vs. reference) and *instruction presence* (with vs. without task description), applied to all 10 available cells in the cross-architecture matrix. OpenVLA additionally provides representations from the DinoSigLIP projector output (4,096-D), DINOv2 and CLIP external baselines, and the branch-split before fusion; these sources support the OpenVLA deep-dive analyses (§4.2).

### 3.3 Cross-Architecture Analyses

Three analyses are applied to every available cell in the $3 \times 4$ matrix.

**Action probes.** To quantify how much action-relevant structure is accessible at each layer, we train two-layer MLP probes [45] to predict 7-DoF action vectors from frozen representations (architecture and training details in Appendix A.2). We report probe $R^2$ as a measure of action decodability. Higher $R^2$ indicates that actions are more accurately predictable from that representation under the probe; it should not be interpreted as evidence that the model causally uses that layer for action generation. We define $\Delta R^2$ as the difference between the maximum layer-wise $R^2$ and the exit-layer $R^2$, and use $\Delta R^2 > 0.10$ as a descriptive threshold for substantial peak-to-exit reduction. Train/test splits follow demonstration boundaries (demo-level splitting reduces $R^2$ from 0.7–0.9 to 0.4–0.6 vs. random splitting; Appendix A.19). 95% CIs are obtained via Monte Carlo cross-validation ($n = 1{,}000$ iterations) with demo-level resampling; the canonical peak layer is fixed from the full-data split to avoid winner's-curse bias. Linear probes yield consistent patterns (Appendix A.14).

**L1 attention head profiling.** To determine whether attention heads preferentially route instruction information to the readout position, each L1 head is classified as instruction-dominant ($> 50\%$ of readout-token attention to instruction tokens) or image-dominant, reported as $k/N$ heads (32 for MHA, 16 for GQA). Sensitivity analysis at thresholds from $30\%$ to $70\%$ confirms ordering robustness (Appendix A.22).

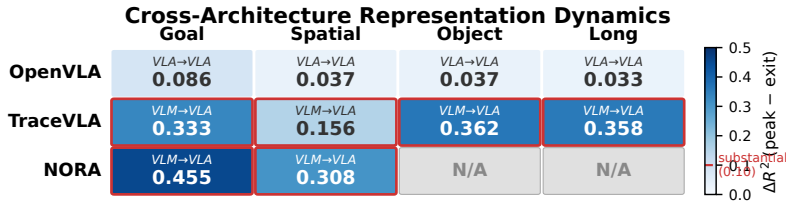
**Counterfactual instruction perturbation.** To test whether the model's actions depend on instruction content, the instruction is replaced with four conditions: empty string, wrong task, shuffled tokens, and paraphrased. We measure trained-model sensitivity via Cohen's $d$ on action MSE (OpenVLA, TraceVLA) or cosine distance (NORA); reference VLMs use cosine distance on hidden representations. These metrics operate on different scales; cross-metric ratios are invalid (§4.4).

OpenVLA additionally supports cross-instruction probes, representational similarity analysis (RSA) [43], and external baselines (DINOv2 [48], CLIP [51]); these single-architecture methods are detailed in Appendix A.2.

## 4 Experiments

### 4.1 Experimental Setup

We apply all three analyses (§3.3) to the $3 \times 4$ architecture-suite matrix described in §3.1 (Table 1). Of the 12 cells, 10 are filled; NORA Object and Long-Horizon lack available checkpoints. For OpenVLA LIBERO-Goal, $N = 63{,}228$ observations; other cells are at comparable scale. For trained models that produce action outputs, counterfactual sensitivity is measured via Cohen's $d$ on action MSE. For reference VLMs (Phi-3-Vision, Qwen2.5-VL) that do not produce actions, sensitivity is measured via cosine distance on hidden representations; these metrics operate on different scales and are not directly comparable (§4.4). OpenVLA's OXE-pretrained checkpoint produces actions, permitting same-metric comparison (Figure 2; Bonferroni correction in Appendix A.30).




Values: peak−exit $\Delta R^2$ from MCCV bootstrap (1000 iterations). Gray cells: suite not available for this architecture.

Figure 2: **Cross-architecture evaluation matrix.** Each cell reports $\Delta R^2$ (peak−exit). OpenVLA: VLA-to-VLA; TraceVLA and NORA: VLM-to-VLA. Gray: unavailable checkpoints. Red borders: $\Delta R^2 > 0.10$.

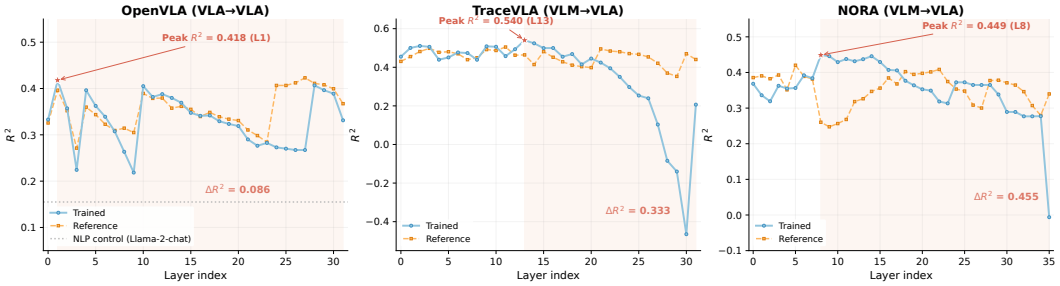



Figure 3: **Layer-wise action-probe $R^2$ (LIBERO-Goal, readout token).** $x$-axis: transformer layer (L0–L31); $y$-axis: action-probe $R^2$. Solid: trained; dashed: reference. VLM-to-VLA: pronounced peak-to-exit drops ($\Delta R^2 = 0.16$–$0.46$); VLA-to-VLA: flat ($< 0.09$). Values use demo-level splitting; see Table 1.

### 4.2 OpenVLA: Motivating Single-Architecture Analysis

Before testing cross-architectural generality, we establish three core phenomena on OpenVLA [2] (full analysis in Appendix A.3). At L1, three conditions form a gradient: random-weight initialization allocates 0/32 instruction-dominant heads, Llama-2 base [49] allocates 9/32, and VLA-trained allocates 24/32, indicating that VLA fine-tuning is associated with instruction routing beyond language-model priors. Counterfactual perturbation produces graded action divergence ordered by semantic distance ($d = 0.81$–$1.18$), and dose-response ablation of instruction-dominant heads yields monotonic trajectory divergence ($d = 0.48$–$0.64$; Appendix A.12). $R^2$ probing localizes training effects at the readout token ($+0.069$ vs. $+0.015$ at image_mean), and readout instruction classification remains at $100\%$ through L31 in the trained model (reference: $82\%$ decay). These phenomena motivate the cross-architecture tests that follow.

### 4.3 Exit Collapse: Layer-Wise Action Probing

We define $R^2$ exit collapse as $\Delta R^2 = R^2_{\text{peak}} - R^2_{\text{exit}}$, the drop in action-probe decodability from peak layer to exit layer (Table 1). VLM-to-VLA fine-tuning is associated with substantial exit collapse across all tested conditions: TraceVLA [7] $\Delta R^2 = 0.156$–$0.362$ across all four suites, and NORA [8] $\Delta R^2 = 0.308$ (Spatial) and $0.455$ (Goal) (Figure 3), all exceeding the $\Delta R^2 > 0.10$ threshold for substantive collapse (§3.3). Under VLA-to-VLA conditions, OpenVLA [2] shows only small fluctuations across all four suites ($\Delta R^2 = 0.033$–$0.086$, all below $0.10$); reference baselines similarly show $\Delta R^2 \leq 0.097$. The gap between the largest non-collapse value ($0.097$) and smallest collapse value ($0.156$) spans $0.059$; any threshold in this interval yields identical grouping, and a permutation test rejects random label assignment ($p = 0.008$; Appendix A.19). Even excluding NORA-Goal (whose bootstrap CI is unreliable; Table 1, footnote $b$), the remaining VLM-to-VLA values span $\Delta R^2 = 0.156$–$0.362$, consistently exceeding the $0.10$ threshold. Monte Carlo cross-validation supports this separation: VLM-to-VLA trained CIs lie entirely above zero (lower bounds $\geq 0.09$), while VLA-to-VLA trained CIs span or nearly span zero (Table 1).

**NLP finetuning control.** To test whether exit collapse is specific to action-oriented training, we probe three chat-finetuned models that share backbones with our VLAs but lack action heads

Table 1: **Peak-to-exit action-probe drop across the architecture-suite matrix.** $\Delta R^2$ is computed as the difference between the maximum layer-wise action-probe $R^2$ and the exit-layer $R^2$. Values above 0.10 are treated as substantial descriptive drops. OpenVLA is a VLA-to-VLA comparison, whereas TraceVLA and NORA are VLM-to-VLA comparisons.

| | | | $\Delta R^2$ (peak − exit) | | | |
|---|---|---|---|---|---|---|
| **Architecture** | **Suite** | **Transition** | **Trained** | **95% CI** | **Reference** | **95% CI** |
| OpenVLA | Goal | VLA→VLA | 0.086 | [−0.060, 0.110] | 0.056 | [0.010, 0.080] |
| OpenVLA | Spatial | VLA→VLA | 0.037 | [0.010, 0.100] | 0.044 | [0.030, 0.110] |
| OpenVLA | Object | VLA→VLA | 0.037 | [0.000, 0.090] | 0.051 | [0.000, 0.090] |
| OpenVLA | Long | VLA→VLA | 0.033 | [0.000, 0.100] | 0.035 | [0.030, 0.100] |
| TraceVLA | Goal | VLM→VLA | 0.333 | [0.210, 0.450] | 0.065 | [−0.110, 0.050] |
| TraceVLA | Spatial | VLM→VLA | 0.156 | [0.090, 0.220] | 0.048 | —[a] |
| TraceVLA | Object | VLM→VLA | 0.362 | [0.280, 0.500] | 0.009 | [−0.070, 0.020] |
| TraceVLA | Long | VLM→VLA | 0.358 | [0.170, 0.530] | 0.097 | [−0.020, 0.110] |
| NORA | Goal | VLM→VLA | 0.455[b] | —[b] | 0.081 | [−0.010, 0.110] |
| NORA | Spatial | VLM→VLA | 0.308 | [0.300, 0.510] | 0.089 | [−0.010, 0.090] |

NORA Object and Long-Horizon checkpoints are unavailable.
[a]CI artifact from flat $R^2$ profile. [b]Bootstrap CI unreliable (exit-layer $R^2 \approx -0.006$).

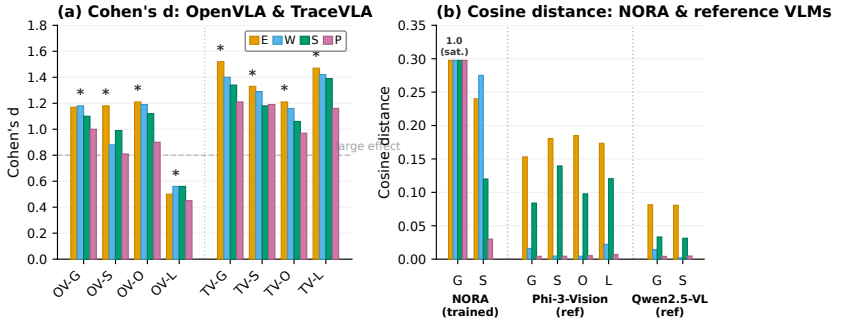



Figure 4: Counterfactual sensitivity across architecture-suite combinations. (a) Cohen's $d$ on action MSE for OpenVLA and TraceVLA; dashed line: $d$=0.8 (large effect). Conditions: E=empty, W=wrong, S=shuffled, P=paraphrased. (b) Cosine distance on hidden representations for NORA and reference VLMs (different scale; §4.4).

or robot data: Llama-2-7b-chat (OpenVLA), Phi-3-mini-4k-instruct (TraceVLA), and Qwen2.5-3B-Instruct (NORA).[1] All three produce flat $R^2$ profiles ($\Delta R^2 = 0.000$; $R^2 = 0.155 \pm 0.001$, $0.155 \pm 0.000$, and $0.155 \pm 0.000$ respectively; Appendix A.36, Figure 24), contrasting with VLM-to-VLA collapse ($\Delta R^2 = 0.156$–$0.455$). The absence of exit collapse in any NLP control is inconsistent with exit collapse being a generic consequence of language-model fine-tuning. **In sum, exit collapse is consistently associated with VLM-to-VLA transitions and absent in both VLA-to-VLA and NLP-only fine-tuning.**

### 4.4 Instruction Sensitivity: Counterfactual Perturbation

**Instruction Sensitivity in Action-Producing VLAs** We first evaluate counterfactual sensitivity in action-producing models using a shared action-space metric. Across OpenVLA and TraceVLA, instruction perturbations produce medium-to-large action divergence in all eight architecture-suite cells (Figure 4).

Every perturbation condition yields $d = 0.45$–$1.52$ (Table 2; full per-condition rankings in Appendix Table 4). The "empty" condition ranks first in 6 of 8 cells ($d = 1.17$–$1.52$); "paraphrased"

[1]Qwen2.5-3B-Instruct shares the Qwen2.5 language backbone with NORA's Qwen2.5-VL-3B but lacks the vision encoder and cross-attention modules; see Appendix A.36.

Table 2: **Counterfactual sensitivity summary.** Cohen's $d$ range and most sensitive perturbation for action-producing models (OpenVLA, TraceVLA). NORA uses cosine distance on representations (different scale; §4.4).

| Architecture | $d$ range | Most sensitive | Metric |
|---|---|---|---|
| OpenVLA | 0.45–1.21 | empty (2/4) | Cohen's $d$ |
| TraceVLA | 0.97–1.52 | empty (4/4) | Cohen's $d$ |
| NORA | 0.28–sat. | — | cosine dist.[a] |

[a] Representation-space only; not comparable to Cohen's $d$.
Empty ranks first in 6/8 cells across both architectures.

ranks last in 7 of 8, indicating that VLAs encode instruction content beyond token-level matching. TraceVLA produces systematically higher effect sizes ($d = 0.97$–$1.52$) than OpenVLA ($0.45$–$1.21$). Removing the instruction entirely often produces larger divergence than replacing it with a wrong task, consistent with instruction removal producing larger disruption than instruction substitution.

Two OpenVLA cells permit same-metric trained/reference comparison: on Spatial, training increases sensitivity (trained−reference difference: $+0.05$ to $+0.24$ across conditions); on Long-Horizon, the OXE-pretrained model exceeds the LIBERO-trained model ($d = 1.05$ vs. $0.57$), consistent with the general-purpose model maintaining broader instruction responsiveness. **Instruction perturbation produces graded action divergence across all tested conditions, with effect sizes reflecting semantic distance.**

**Representation-Space Sensitivity in NORA and Reference VLMs** NORA and reference VLMs lack action outputs; their sensitivity is measured via cosine distance on hidden representations (different scale from Cohen's $d$). NORA Goal produces saturated cosine distances ($1.0$); NORA Spatial peaks at $0.275$; reference VLMs show weak sensitivity ($\leq 0.19$; Appendix A.4). The trained-reference contrast is consistent with action-space findings, though the different metric prevents direct magnitude comparison.

### 4.5 Inherited Routing: L1 Attention Profiling

L1 instruction-dominant head counts across architectures reveal a pattern that contrasts with the training-associated effects documented in §4.3 and §4.4. The TraceVLA reference (Phi-3-Vision) allocates 32/32 heads to instruction-dominant attention across all four suites, saturating the capacity of its 32-head L1 before any VLA training. Training slightly reduces this count to 27–30/32, indicating that VLA fine-tuning does not increase L1 instruction-dominant attention in this architecture. The NORA reference (Qwen2.5-VL) allocates 13–14 of its 16 L1 heads to instruction dominance (81–88%), and training leaves this essentially unchanged at 12–13/16. OpenVLA shows 24–26/32 instruction-dominant heads in the OXE-pretrained baseline; trained values are similar (14–24/32), with the largest change in Long-Horizon where training reduces the count from 25 to 14 heads (Figure 5).

If VLA training drove L1 instruction routing, reference models should show low counts that increase after training, as in OpenVLA's random-weight condition (0/32; §4.2); instead, both reference VLMs already saturate or near-saturate, and training leaves counts unchanged or slightly reduced. **L1 instruction routing is an inherited backbone property, not a consequence of VLA training.**

### 4.6 Synthesis: Training-Associated vs. Inherited Routing

The three analyses separate two empirical patterns: exit collapse and counterfactual sensitivity are training-associated (pronounced in VLM-to-VLA, small in VLA-to-VLA), while L1 instruction attention is inherited from reference backbones. Despite qualitatively different reference starting

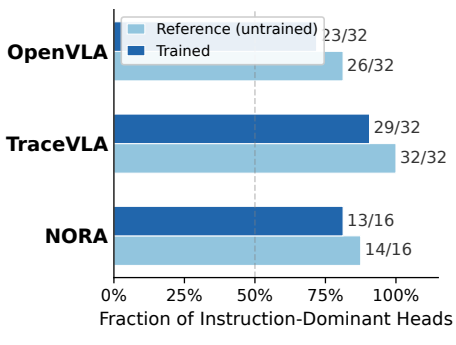



Figure 5: L1 instruction-dominant head

points (OpenVLA: peak at L0 with monotonic decline; Phi-3-Vision: gradual decline from L6), trained OpenVLA and TraceVLA converge to nearly identical mid-layer peaks ($R^2 = 0.637$ at L9 vs. $0.634$ at L11, $\Delta = 0.003$; Appendix A.29, Table 3), suggesting that action fine-tuning is associated with convergence toward a shared mid-layer structure regardless of starting point. These probes measure decodability, not causal reliance (label-permutation controls confirm genuine structure; Appendix A.19). Matched-size ablation reveals that instruction sensitivity is distributed across heads in proportion to their instruction attention fraction, consistent with instruction information being woven into the general residual stream rather than concentrated in a dedicated routing circuit ($d = 0.65$ vs. $0.60$, 8% gap; Appendix Table 26). This distributed pattern suggests that uniform pruning may be preferable to attention-score-based head selection for VLA compression. The consistent three-way pattern provides a first cross-architecture decomposition of VLA instruction routing.

# 5 Discussion and Conclusion

**Scope and limitations.** All findings are correlational (confounded by data distribution and optimization; $N = 2$ VLM-to-VLA transitions; Appendix A.37); we analyze causal-decoder VLAs on LIBERO only (flow-matching [14] and diffusion [52] architectures may differ); all analyses are open-loop. Trained $\Delta R^2$ correlates with published success rates ($\rho = 0.583$, $p = 0.10$; Appendix A.33) but is underpowered ($n = 9$; effect size is large, $|\rho| > 0.5$); tracing collapse onset remains for future work.

**Practical implications and conclusion.** $\Delta R^2 > 0.10$ provides a cheap probe-based screening signal that can supplement rollout evaluation. The flat VLA-to-VLA $\Delta R^2$ profile justifies staged pipelines that preserve exit structure, and mid-layer features (peak at L9–L11) outperform exit-layer representations where collapse is pronounced. The distributed instruction integration pattern (§4.6) implies that uniform pruning may be preferable to attention-score-based head selection for VLA compression. The consistent three-way pattern provides a first cross-architecture decomposition of VLA instruction routing; extended implications are discussed in Appendix A.34. Code and data will be released upon acceptance.

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**Use of Generative AI.** Generative AI tools (Claude, GPT-4) assisted with code implementation and manuscript editing; all scientific content was produced by the authors.

# A Supplementary Material

Unless noted otherwise, all analyses use OpenVLA-7B (`openvla/openvla-7b` on HuggingFace) evaluated on LIBERO-Goal (10 tasks $\times$ 50 demonstrations, $N = 63{,}228$ observations after excluding terminal frames). Model conditions, representation extraction, metric definitions, and probe architectures follow §3. Cross-suite experiments (§A.18) use LIBERO-Spatial with the same protocol (10 tasks $\times$ 50 demonstrations). All experiments ran on 2$\times$ NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs (96 GB each); random seed is 42 unless stated otherwise.

## A.1 Architecture Summary

Table 3: Architectures studied. The reference baselines differ in kind: the OpenVLA reference is an OXE-pretrained VLA that produces actions; TraceVLA and NORA references are pure VLMs with no action training.

| Arch. | Backbone | Layers | Attention | Vision Enc. | Reference Baseline |
|---|---|---|---|---|---|
| OpenVLA | Llama-2-7B | 32 | 32 MHA | DinoSigLIP $\rightarrow$ MLP proj. | OXE-pret. VLA |
| TraceVLA | Phi-3-mini | 32 | 32 MHA | SigLIP + spatial trace | Phi-3-Vision (VLM) |
| NORA | Qwen2.5-VL-3B | 36 | 16q/2kv GQA | Qwen2.5-VL native | Qwen2.5-VL (VLM) |

## A.2 OpenVLA-Specific Analysis Methods

Three additional analyses are applied to OpenVLA to characterize instruction encoding and representational geometry.

**Cross-instruction probes.** Each visual frame is paired with all 10 LIBERO-Goal task instructions, producing 10 representation vectors per frame per layer. A 10-way linear classifier identifies which instruction was present, applied separately at image_mean and readout, with leave-one-out cross-validation (LOOCV) over 10 folds.

**Representational similarity analysis (RSA).** RSA [43] quantifies geometric alignment between representations and the action space. For each task, we compute pairwise representational dissimilarity matrices (RDMs) using cosine distances for representations and Euclidean distances for 7-D action vectors, from within-task observation pairs across 50 demonstrations. Alignment is measured via Spearman $\rho$ between upper-triangular RDM entries; we report mean $\rho$ across 10 tasks with 95% confidence intervals from 10,000 bootstrap resamples at the task level [53]. We prefer RSA over

CKA [44]: RSA captures relative distance structure via scale-invariant rank correlation, suited to comparing high- and low-dimensional spaces (4,096-D vs. 7-D), while CKA measures global subspace overlap that can remain high when geometric structure differs. Both metrics are reported in §A.5; their divergence (§4.6) confirms that neither alone fully characterizes representational structure.

**External baselines.** DINOv2 ViT-L/14 [48] (`cls_token`/`patch_mean`) and CLIP ViT-L/14 [51] (`patch_mean`) provide self-supervised and vision-language pretrained reference points for probing and RSA.

### A.3 OpenVLA Single-Architecture Deep Dive

This section provides the full single-architecture analysis summarized in §4.2.

**Cross-instruction probes.** Cross-instruction probes validate the analysis framework on OpenVLA [2]: as predicted by the causal mask, image_mean instruction classification stays at chance ($10\%$) across all 32 layers in all conditions (full profile in §A.22). Trained readout achieves $100\%$ 10-way classification from L1 onward and maintains this through L31 (LOOCV-confirmed; §A.22). Reference readout also reaches $100\%$ at L1 but decays to $82\%$ by L31; the readout token is thus the primary multimodal integration site, and instruction encoding remains stable across depth in the trained model. All patterns reproduce with a linear probe (§A.14).

**L1 attention gradient.** At L1, three conditions form a gradient of instruction attention: randomly initialized networks allocate 0/32 instruction-dominant heads ($11.7\%$ text attention, $1.5\times$ text-to-image ratio), Llama-2 base [49] allocates 9/32 ($36.5\%$), and VLA-trained allocates 24/32 ($57.7\%$, $38.2\times$ text-to-image ratio; §A.23). The 32-layer attention profile reveals qualitatively distinct organizations: Llama-2 base concentrates instruction attention at mid-depth (L9–L16: 15–22 heads), while VLA fine-tuning produces an entry-exit bookend pattern with L0–L1 at 21–24 heads, L31 at 18, and intermediate layers at 0–11 (§A.22). Head counts are computed at a $50\%$ threshold; sensitivity analysis ($30\%$–$70\%$) confirms ordering robustness. A fourth condition, OpenVLA fine-tuned on LIBERO, preserves L1 entry concentration (22/32 heads) but loses L31 exit re-engagement entirely (0 vs. 18 heads for the OXE-pretrained model), consistent with exit-layer attention serving multi-task flexibility rather than task-specific routing. The redistribution beyond Llama-2 base priors is consistent with VLA fine-tuning being associated with instruction routing beyond what is inherited from language-model attention.

**Counterfactual perturbation.** Four perturbation types across 10 LIBERO-Goal tasks produce large action divergence: wrong task ($d = 1.182$), empty instruction ($1.170$), shuffled tokens ($1.096$), paraphrased ($0.996$). All four conditions yield $d \geq 1.0$, with "wrong" slightly exceeding "empty." The effect replicates across LIBERO-Spatial (10 tasks): empty $1.173$, shuffled $0.979$, wrong $0.885$, paraphrased $0.807$ (Table 9). Across both suites (20 tasks), the gradient spans $d = 0.81$–$1.18$. Spatial instructions differ only in prepositions, yet wrong-task perturbation still produces $d = 0.885$; instruction-dominant heads route fine-grained spatial information, not merely task identity.

**Dose-response ablation.** Dose-response ablation of L1 instruction-dominant heads produces monotonic trajectory divergence: $d$ rises from $0.482$ to $0.639$ as 4–32 heads are progressively ablated, with steep increase through 12 heads and plateau beyond 20 (§A.12). Ablating all 24 instruction-dominant heads alters $72.3\%$ of action tokens (token-level $d = 0.65$, trajectory-level $d = 2.943$); control ablation of 8 random heads yields $d = 0.41$ ($29.3\%$ tokens affected), concentrated on the gripper dimension. The routing effect is task-dependent: instruction-discriminative tasks show $2.5\times$ larger $|\Delta R^2|$ under L1 ablation than visually-sufficient tasks ($p = 0.033$; §A.26).

**$R^2$ localization.** $R^2$ action probing localizes training effects: trained readout peaks at $R^2 = 0.678$ (L13), reference readout peaks at $0.609$ (L32), and the training effect concentrates at the readout

(+0.069) rather than at image_mean (+0.015; §A.10). These effects survive Bonferroni correction across all 63 tests (§A.30).

### A.4 Counterfactual Sensitivity Details

Table 4 reports per-condition counterfactual rankings for all eight cells where Cohen's $d$ is available.

Table 4: Counterfactual condition ranking for trained models (Cohen's $d$; higher = greater action divergence under instruction perturbation). Conditions: E = empty instruction, W = wrong task, S = shuffled tokens, P = paraphrased. "Empty" ranks first in 6 of 8 cells; OpenVLA Goal ("wrong" first) and Long ("shuffled" first) are the exceptions. NORA cells use cosine distance and are reported in §4.4.

| Arch. | Suite | 1st | 2nd | 3rd | 4th |
|---|---|---|---|---|---|
| OpenVLA | Goal | W 1.18 | E 1.17 | S 1.10 | P 1.00 |
| OpenVLA | Spatial | E 1.17 | S 0.98 | W 0.89 | P 0.81 |
| OpenVLA | Object | E 1.21 | W 1.19 | S 1.12 | P 0.90 |
| OpenVLA | Long | S 0.57 | W 0.55 | E 0.51 | P 0.45 |
| TraceVLA | Goal | E 1.52 | W 1.40 | S 1.34 | P 1.21 |
| TraceVLA | Spatial | E 1.33 | S 1.30 | P 1.18 | W 1.17 |
| TraceVLA | Object | E 1.21 | S 1.16 | W 1.06 | P 0.97 |
| TraceVLA | Long | E 1.47 | W 1.42 | S 1.39 | P 1.16 |

**Metric constraints on trained–reference comparison.** Trained models produce action outputs, enabling Cohen's $d$ on action MSE (OpenVLA, TraceVLA) or cosine distance on action tokens (NORA). Reference VLMs (Phi-3-Vision, Qwen2.5-VL) do not generate actions; their sensitivity is measured via cosine distance on hidden representations—a metric with different scale, units, and distributional properties. Ratios between Cohen's $d$ and cosine distance are therefore methodologically invalid, and we do not report cross-metric amplification factors. Qualitatively, reference VLMs show weak sensitivity: the TraceVLA reference (Phi-3-Vision) produces peak cosine distances of 0.15–0.19; the NORA reference (Qwen2.5-VL) yields 0.081 on Goal.

**NORA details.** NORA Goal produces saturated exit-layer cosine distances of 1.0 for all conditions, with differentiation visible only at earlier layers (L24: wrong = 0.123, empty = 0.098). NORA Spatial produces a peak cosine distance of 0.275 for the "wrong" condition.

### A.5 CKA Analysis Method

We employ Centered Kernel Alignment (CKA) [44] alongside RSA as a complementary perspective. While RSA captures local geometric alignment via rank-order preservation, CKA measures global representational similarity through kernel-based comparisons.

We compute both *linear CKA*, which uses a linear kernel and is sensitive to second-order statistics, and *RDM kernel CKA*, which operates on the same distance matrices as RSA but aggregates them differently. Using both variants tests whether conclusions about representation structure are robust to the choice of similarity metric.

Linear CKA between high-dimensional representations (4,096-D for LLM layers) and the 7-D action space is subject to a known dimension mismatch: the linear kernel in the low-dimensional space may not capture the same structure as in the high-dimensional space. RDM kernel CKA avoids this issue by operating on distance matrices of equal size regardless of the original feature dimensionality.

### A.6 Metric Sensitivity: RSA versus CKA

Linear CKA [44] reverses the RSA ordering: reference LLM (CKA = 0.089) > trained LLM (0.083) > DinoSigLIP+Projector (0.081) > DINOv2 (0.066), partially inverting the self-supervised

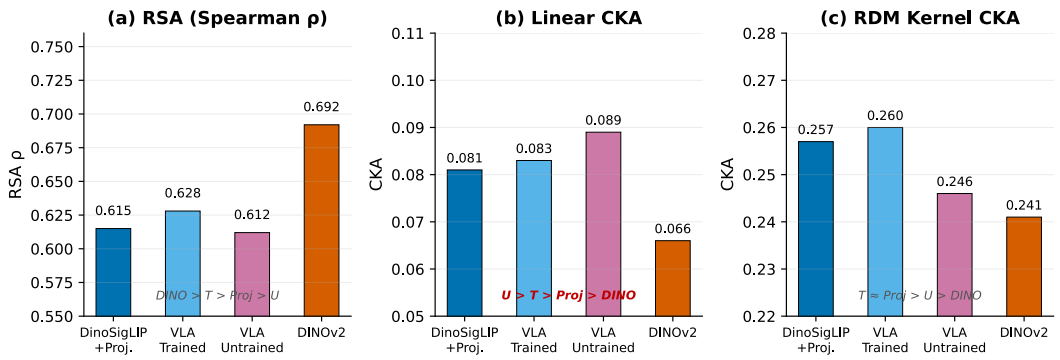



Figure 6: Three similarity metrics across representation sources. **(a)** RSA ranks DINOv2 cls (0.692) > Fused (0.634) > Projector (0.615). **(b)** Linear CKA reverses the ordering: Reference (0.089) > Trained (0.083) > Projector (0.081) > DINOv2 (0.066). **(c)** RDM kernel CKA: Trained (0.260) ≈ Projector (0.257) > Reference (0.246) > DINOv2 (0.241). The metric divergence underscores that RSA and CKA capture different aspects of representational structure.

advantage (Figure 6). RDM kernel CKA partially resolves this discrepancy: trained (0.260) ≈ Projector (0.257) > reference (0.246) > DINOv2 (0.241). This divergence reflects the known sensitivity of representation similarity metrics to different geometric properties.

At the per-layer level, linear CKA between image_mean and the action space rises monotonically from L0 (0.081) to L32 (0.098), with no inverted-U pattern. RSA captures rank-order geometric alignment; CKA captures global linear structure. The dissociation indicates that the geometric refinement in early layers operates on the relative arrangement of representations (affecting RSA) without altering the overall covariance structure (captured by CKA).

### A.7 Branch-Split Detailed Analysis

To disentangle the contributions of the two encoder streams within DinoSigLIP, we extract representations from the DINOv2 branch (1,024-D) and SigLIP branch (1,152-D) separately before fusion. The two branches yield near-identical RSA: DINOv2 branch $\rho = 0.619$, SigLIP branch $\rho = 0.621$. Fusion produces a representation ($\rho = 0.634$) that exceeds either branch alone; concatenation captures complementary structure. The MLP projector then compresses this signal: DinoSigLIP+Projector achieves $\rho = 0.615$, a drop of $\Delta = 0.019$ from the fused 2,176-D input.

### A.8 RSA Across Representation Sources

Figure 7 shows RSA with the action space across all representation sources; the ordering matches the RSA results reported in §4.6.

### A.9 Per-Layer RSA Profiles

Figure 8 extends Figure 3 (main text) by adding readout and reference profiles.

### A.10 Per-Layer Action Decoding Across Training Stages

Figure 9 compares per-layer action probe $R^2$ across three training stages (randomly initialized, Llama-2 base, VLA-trained) using 50 demonstrations per task, averaged over 3 seeds (mean ± std). All three conditions share L0 representations ($R^2 = 0.645 \pm 0.008$); subsequent differences thus arise from Transformer processing rather than input variation. The trained model peaks at L9 ($R^2 = 0.643 \pm 0.014$), consistent with progressive specialization for action-relevant feature integration. Llama-2 base shows a steady decline (L0: $0.645 \rightarrow$ L32: $0.551$, std $0.003$–$0.035$), consistent with stable but unspecialized processing. The randomly initialized model exhibits high cross-seed variance and deep-layer collapse, with negative $R^2$ and bimodal behavior across seeds at late layers;

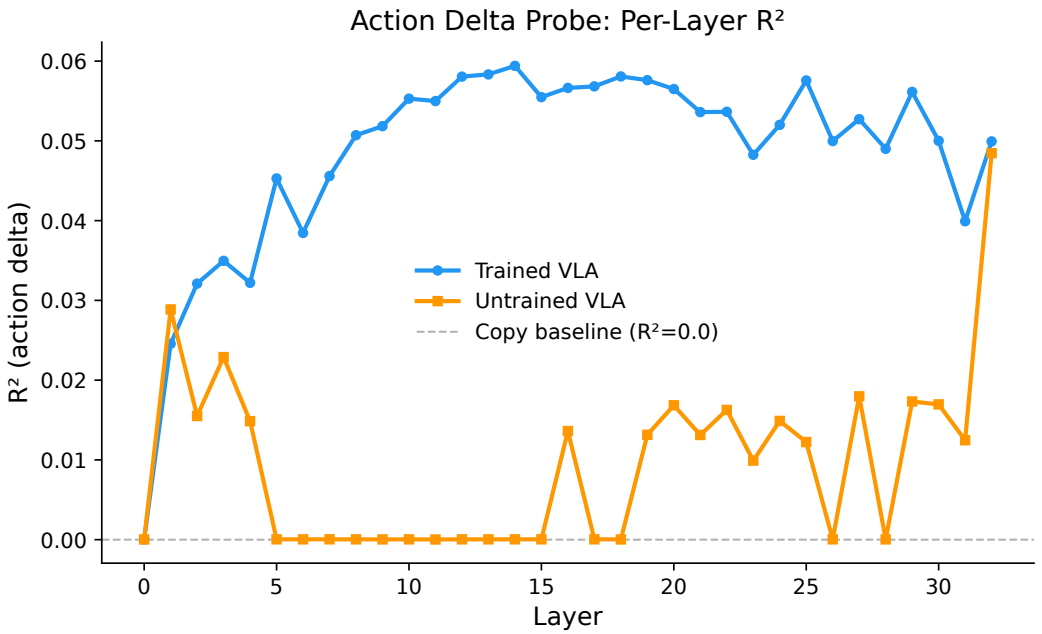



Figure 7: RSA with the action space ($\rho$) across representation sources. Self-supervised DINOv2 `cls_token` ($\rho$=0.692) achieves the highest geometric alignment. CLIP (0.557) yields the weakest alignment among pretrained encoders. Error bars: 95% bootstrap CIs.

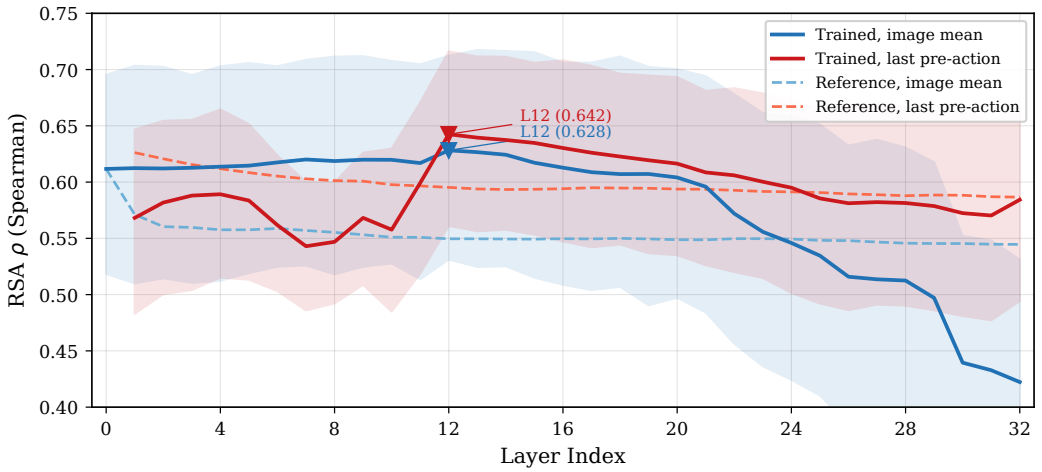



Figure 8: Per-layer RSA ($\rho$) for trained and reference models at image_mean and readout positions. The trained model shows an inverted-U peaking at L12 for both positions; the reference model shows monotonic decline. After L12, readout and image_mean diverge: readout $\rho$ declines gradually ($0.628 \rightarrow 0.584$) while image_mean $\rho$ drops sharply ($0.628 \rightarrow 0.422$), consistent with task-specific conditioning at the readout position.

this instability indicates that randomly initialized Transformer layers produce destructive interference on action-relevant structure. Random weights at intermediate layers produce anti-correlated representations ($R^2 < 0$, L10–L11), consistent with randomly initialized networks lacking structured computation. The three-condition gradient (trained $>$ Llama-2 base $>$ randomly initialized) holds from L1 through L15, with a reversal at L32 where trained $R^2$ drops below the pretrained baseline. The qualitative pattern is consistent across all 3 seeds. Three conditions show substantial agreement in layer-wise $R^2$ ordering (Kendall's $W = 0.72$, $\chi^2 = 68.99$, $p < 0.001$).

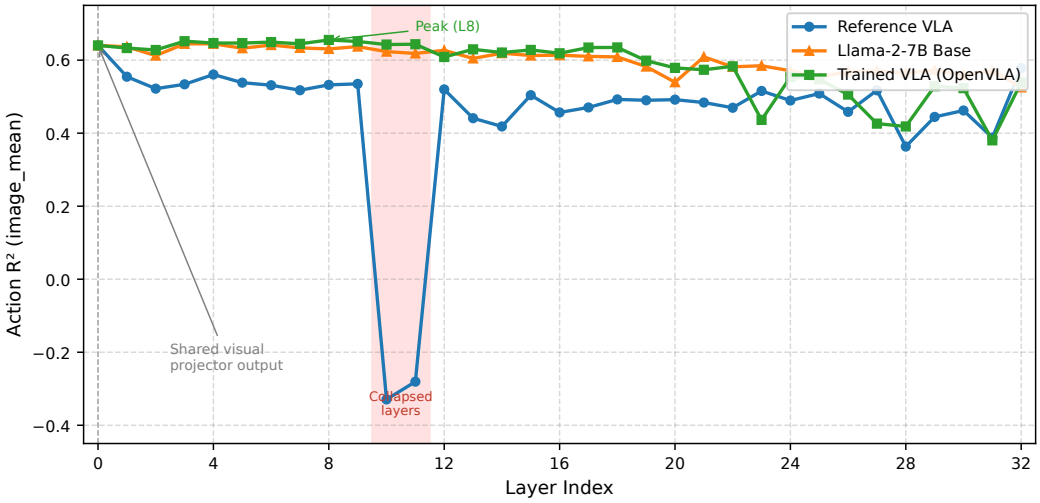



Figure 9: Per-layer action probe $R^2$ (image_mean) across three training conditions: randomly initialized VLA (random weights), Llama-2 base (pretrained, no VLA fine-tuning), and VLA-trained. All conditions share L0 ($0.645 \pm 0.008$). Trained model peaks at L9; Llama-2 base declines steadily; randomly initialized shows high cross-seed variance and deep-layer collapse. 50 demonstrations per task, 3 seeds (mean $\pm$ std).

### A.11 Random-Subset Ablation Details

To evaluate whether the ablation effect in §4.2 is specific to instruction-dominant heads, we compare dominant-head ablation against 30 random subsets of 24/32 L1 heads (each subset drawn without replacement). Dominant-head ablation increases action $R^2$ by $+0.030$ (within-experiment baseline $0.524$), while random subsets yield a mean increase of $+0.071 \pm 0.013$ (Figure 10). The dominant-head effect falls below all 30 random subsets ($z = -3.11$, percentile $= 0\%$); heads with higher instruction-attention fraction carry more organized attention signals, consistent with graded rather than binary specialization.

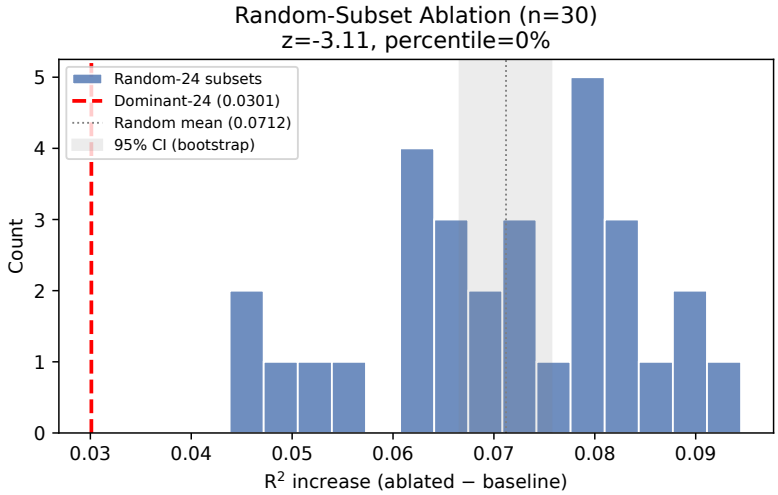



Figure 10: $R^2$ increase from ablating 24/32 L1 heads. Gray bars: 30 random subsets; red line: dominant-24 heads ($+0.030$). Random subsets yield larger gains (mean $+0.071 \pm 0.013$; $z = -3.11$, percentile $= 0\%$), consistent with graded attention specialization across L1 heads.

### A.12 Action Trajectory Analysis Details

Table 8 reports per-dimension mean absolute shift for instruction-dominant (24 heads) and control (8 heads) L1 ablations. Both conditions produce negligible shifts on translational dimensions ($dx$, $dy$, $dz < 0.01$) and small rotational shifts ($rx$, $ry$, $rz < 0.04$). The gripper dimension absorbs the majority of the effect: instruction-head ablation shifts gripper predictions by $0.291$ (control: $0.141$), consistent with L1 attention influencing discrete open/close decisions.

Per-task Cohen's $d$ values for instruction-head ablation are $0.43$, $0.69$, $0.70$, $0.70$, and $0.73$ (mean $0.65$), with all tasks significant ($p < 10^{-10}$). Token-level KL divergence between original and ablated action distributions is $5.08$ for instruction heads and $1.69$ for control heads.

**Trajectory-level aggregation.** Aggregating per-demo mean actions to the trajectory level reveals that token-level $d$ values are conservative: within-trajectory averaging reduces noise, yielding $\sim 4\times$ larger effect sizes (mean token-to-trajectory ratio $0.245$). Instruction-head ablation ($N = 50$: 5 tasks $\times$ 10 demos) yields trajectory $d = 2.943$ (95% CI $[2.469, 3.707]$); bottom-24 non-instruction heads ($N = 30$) yield $d = 2.347$ (CI $[1.854, 3.221]$). All 14 conditions produce trajectory $d > 2.0$ with CI lower bounds $> 1.7$. Counterfactual trajectory-level $d$ (10 LIBERO-Goal tasks, $N = 100$): wrong $3.798$ (CI $[3.194, 4.938]$), empty $3.438$ ($[2.921, 4.448]$), shuffled $2.595$ ($[2.028, 3.791]$), paraphrased $2.131$ ($[1.740, 2.819]$). Token-level $d$ underestimates trajectory-level effects by $\times 3.1$–$4.5$, consistent with within-trajectory averaging reducing noise. Fine-grained ordering is not reliably distinguished at trajectory level ($N = 100$); token-level comparisons ($N \sim 5{,}000$) provide higher resolution for ordering claims.

**Ground-truth action error.** Ablation redirects model outputs ($d = 0.65$) but does not systematically increase error relative to expert demonstrations: original MSE $= 0.498$; top-24 instruction-head ablation MSE $= 0.497$ ($\Delta = -0.001$, Wilcoxon $p = 0.52$, n.s.); bottom-24 non-instruction heads MSE $= 0.506$ ($\Delta = +0.008$). Changes concentrate on the gripper dimension; translation and rotation dimensions are negligibly affected. Per-task breakdown shows all $|\Delta\text{MSE}| < 3\%$ of baseline with all $p > 0.27$; the near-zero aggregate $\Delta$ reflects uniformly small per-task effects rather than cancellation across tasks. This is consistent with the information-computation dissociation: instruction-dominant heads control instruction-specific direction selection rather than action precision.

Table 5 reports per-head bootstrap CIs ($10{,}000$ resamples). Individual control heads produce $\sim 1.9\times$ stronger effects ($d = 0.051$) than instruction heads ($d = 0.027$, $p = 0.0018$), consistent across all 5 tasks. The 24 instruction heads nonetheless produce a larger aggregate perturbation ($d = 0.65$ vs. $0.41$), consistent with collective coordination rather than individual strength.

Table 5: Per-head bootstrap analysis of L1 ablation effects. Instruction heads are individually weaker but collectively stronger, consistent with coordinated attention redistribution.

| **Group** | $N$ **heads** | **Collective** $d$ | **Per-head** $d$ **[95% CI]** |
|---|---|---|---|
| Instruction-dominant | 24 | 0.65 | 0.027 [0.026, 0.029] |
| Control | 8 | 0.41 | 0.051 [0.048, 0.055] |
| Difference | — | — | 0.024, $p = 0.0018$ |

Instruction-head ablation produces more uniform perturbation across action dimensions (CV $= 1.30$ vs. control CV $= 2.74$, permutation $p = 0.016$), consistent with structured, language-conditioned computation rather than diffuse noise.

Table 6 and Figure 11 report the dose-response gradient for instruction-head ablation at L1. Trajectory divergence increases monotonically from $d = 0.482$ (4 heads) to $d = 0.639$ (32 heads), with steep rise through 12 heads (slope $\Delta d / \Delta 4 \approx 0.026$), a plateau at 16–20 heads ($\Delta d = 0.001$), and slow increase at 20–32.

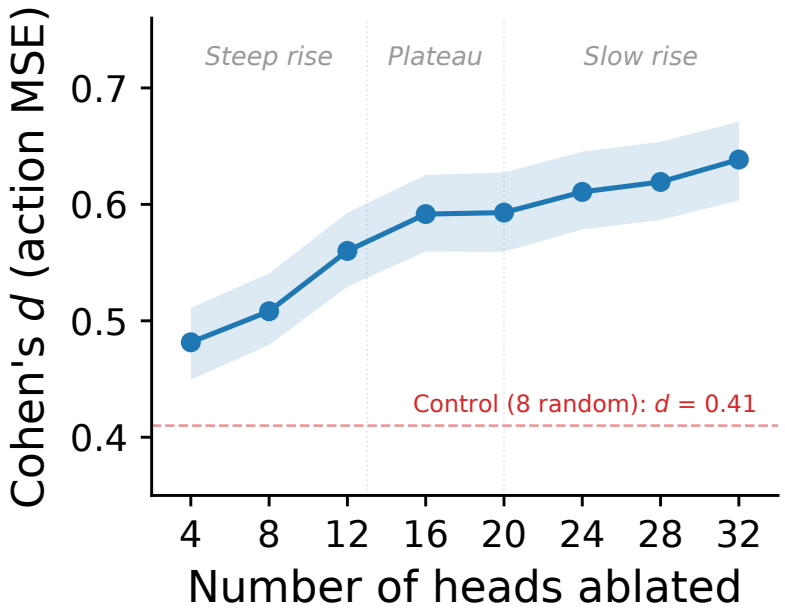



Figure 11: Dose-response gradient: trajectory divergence (Cohen's $d$) as a function of ablated instruction-dominant heads at L1. Error bars: 95% bootstrap CIs (10,000 resamples). Dashed red: control ablation (8 non-instruction heads, $d = 0.41$). Steep rise through 12 heads, plateau at 16–20, diminishing returns beyond.

Table 6: Dose-response data for L1 instruction-head ablation. Cohen's $d$ vs. original action sequence; 95% bootstrap CIs (10,000 resamples).

| **Heads ablated** | **Cohen's $d$** | **95% CI** |
|---|---|---|
| 4 | 0.482 | [0.449, 0.511] |
| 8 | 0.508 | [0.479, 0.541] |
| 12 | 0.560 | [0.529, 0.593] |
| 16 | 0.592 | [0.559, 0.625] |
| 20 | 0.593 | [0.559, 0.627] |
| 24 | 0.611 | [0.578, 0.645] |
| 28 | 0.619 | [0.587, 0.654] |
| 32 | 0.639 | [0.603, 0.671] |

### A.13 Pretrained LLM Representation Quality Under Visual Input

A potential concern with the three-condition comparison (§3.1) is that the pretrained Llama-2 backbone never encountered projected visual tokens during pretraining, so its representations of such input could be degenerate. We evaluate three diagnostics to assess representation quality. Activation magnitude grows from 16.5 (L0) to 90.2 (L31) for Llama-2 base, compared to 16.5 $\rightarrow$ 51.7 for the VLA-trained model; the pretrained LLM produces larger but non-exploding activations with no evidence of collapse. Per-layer cosine similarity between L0 and deeper layers decreases progressively from 1.0 to $\approx 0.28$ at L25, consistent with gradual differentiation rather than catastrophic out-of-distribution behavior. Top-10 principal component variance explained remains stable across all layers (0.59–0.74), with no low-rank collapse. These diagnostics indicate that the pretrained LLM produces structured, full-rank representations of visual input despite never encountering projected visual tokens during pretraining (Figure 12). These results support the validity of the three-condition comparison.

Table 7: Counterfactual instruction analysis across all 10 LIBERO-Goal tasks. Jaccard: token overlap with original instruction; Cohen's $d$: mean across 10 tasks. "Wrong" slightly exceeds "empty" ($d = 1.182$ vs. 1.170); all four conditions produce large effects ($d \geq 1.0$). L1 ablation included for comparison.

| **Condition** | **Jaccard** | **Cohen's $d$** |
|---|---|---|
| Empty (no instruction) | 0.000 | 1.170 |
| Wrong (different task) | 0.174 | 1.182 |
| Shuffled (random order) | 1.000 | 1.096 |
| Paraphrased | 0.533 | 0.996 |
| L1 ablation (24 heads) | — | 0.650 |

Table 8: Mean absolute action shift per dimension under L1 head ablation. Instruction: 24 dominant heads; Control: 8 remaining heads. Gripper shift dominates both conditions.

| **Condition** | $dx$ | $dy$ | $dz$ | $rx$ | $ry$ | $rz$ | **Gripper** |
|---|---|---|---|---|---|---|---|
| Instruction (24) | 0.002 | 0.005 | 0.008 | 0.021 | 0.031 | 0.034 | 0.291 |
| Control (8) | 0.000 | 0.001 | 0.002 | 0.008 | 0.014 | 0.018 | 0.141 |

### A.14 Probe Architecture Sensitivity

To rule out that our findings depend on MLP probe capacity ($N/\text{params} \approx 0.06$, overparameterized), we replicate the per-layer analysis with a linear probe ($4{,}096 \to 7$, $N/\text{params} \approx 2.2$). The inverted-U pattern reproduces: trained image_mean peaks at L14 ($R^2 = 0.565$), while the reference declines monotonically from L0 ($R^2 = 0.517$). The trained–reference gap is larger under the linear probe (L14: $\Delta = +0.324$) than the MLP probe (L8: $\Delta = +0.015$); trained representations are thus more probe-decodable for action information. Reference deep layers collapse entirely under the linear probe (L25: $R^2 = -0.298$); MLP nonlinearity partially rescues these degraded representations. For readout tokens, the linear probe also corroborates the inverted-U (peak L13, $R^2 = 0.596$), with the same peak layer as the MLP probe and a trained-vs-reference gap of $\Delta = 0.103$ ($p < 0.001$). These results indicate that our qualitative findings reflect representational structure rather than probe capacity artifacts (Figure 13).

### A.15 Functional Probe Comparison

Figure 14 compares functional probe $R^2$ across representation sources at each source's best layer.

### A.16 Training Effect and Instruction Invariance

Figure 15 shows trained vs. reference RSA profiles and instruction invariance. Instruction invariance holds across all conditions: maximum $|\Delta| < 0.006$, so action geometry is visually grounded and not modulated by language instructions.

### A.17 Additional RSA Values

DINOv2 `patch_mean` achieves RSA $\rho = 0.631$, between the `cls_token` (0.692) and the fused DinoSigLIP (0.634). VLA reference readout peaks at L1 ($\rho = 0.626$).

### A.18 Cross-Suite Validation within LIBERO: Spatial

To assess generalization beyond in-distribution tasks, we repeat the core RSA and probe analyses on LIBERO-Spatial (10 out-of-distribution tasks $\times$ 50 demonstrations each, requiring spatial reasoning). Figure 16 shows per-layer RSA and $R^2$ profiles for both LIBERO-Goal and LIBERO-Spatial side by side.

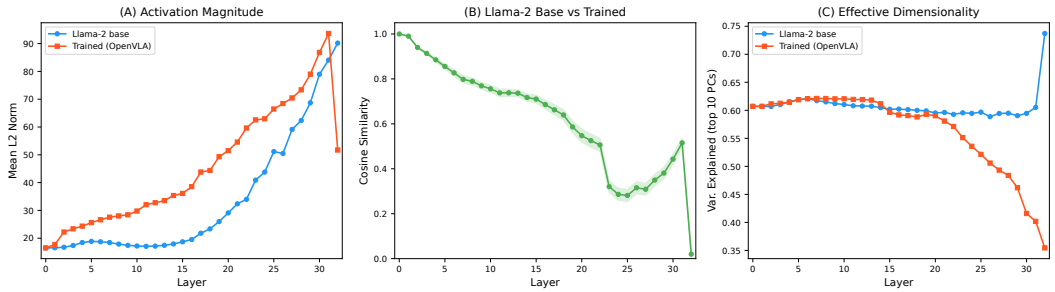



Figure 12: Pretrained LLM representation diagnostics under visual input. Left: activation magnitude across layers (Llama-2 base vs. VLA-trained). Center: cosine similarity between L0 and deeper layers. Right: top-10 PC variance explained. All three metrics indicate structured, non-degenerate representations.

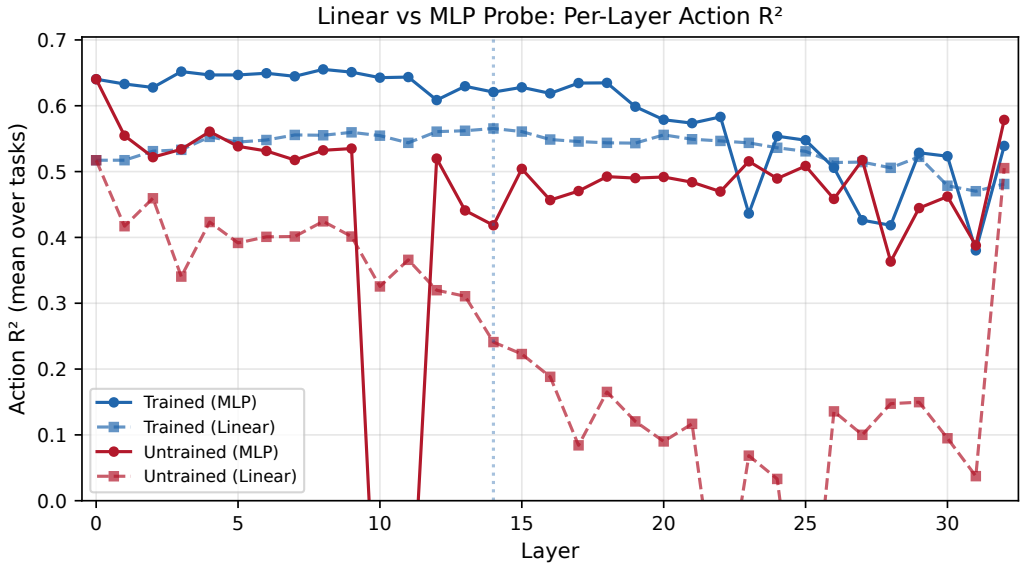



Figure 13: Per-layer action probe $R^2$ for linear (dashed) and MLP (solid) probes across trained (blue) and reference (red) conditions. The inverted-U reproduces under both probe architectures; the trained–reference gap is larger for linear probes.

The inverted-U profile replicates on LIBERO-Spatial: trained image_mean RSA peaks at L14 ($\rho = 0.691$) and $R^2$ at L13 (0.696), with peaks shifting $\sim$2 layers deeper than LIBERO-Goal (RSA L12, $R^2$ L8–L10), consistent with increased task difficulty requiring deeper processing. Training produces a significant geometric gain (best-vs-best $\Delta = +0.023$, 95% CI $[0.012, 0.036]$, $p < 0.001$), larger than the LIBERO-Goal effect ($\Delta = +0.017$, CI includes zero).

Two notable differences are observed. First, the absolute RSA values are higher on Spatial (0.691 vs. 0.628), possibly reflecting greater action diversity in spatial tasks that produces a more discriminative action RDM. Second, RSA–$R^2$ peak convergence does not replicate: readout RSA peaks at L15 while $R^2$ peaks at L21 (6-layer gap), contrasting with the near-alignment on Goal ($|\Delta_{\text{layer}}| = 1$). This suggests that peak convergence is benchmark-specific rather than a general consequence of training.

**Cross-Suite Counterfactual Replication** Table 9 reports counterfactual instruction analysis on LIBERO-Spatial (10 tasks). The ordering differs from LIBERO-Goal: empty > shuffled > wrong > paraphrased (vs. wrong > empty > shuffled > paraphrased on Goal). All global effect sizes exceed $d = 0.8$ (per-task wrong minimum is 0.591). Spatial instructions differ only in prepositions (e.g.,

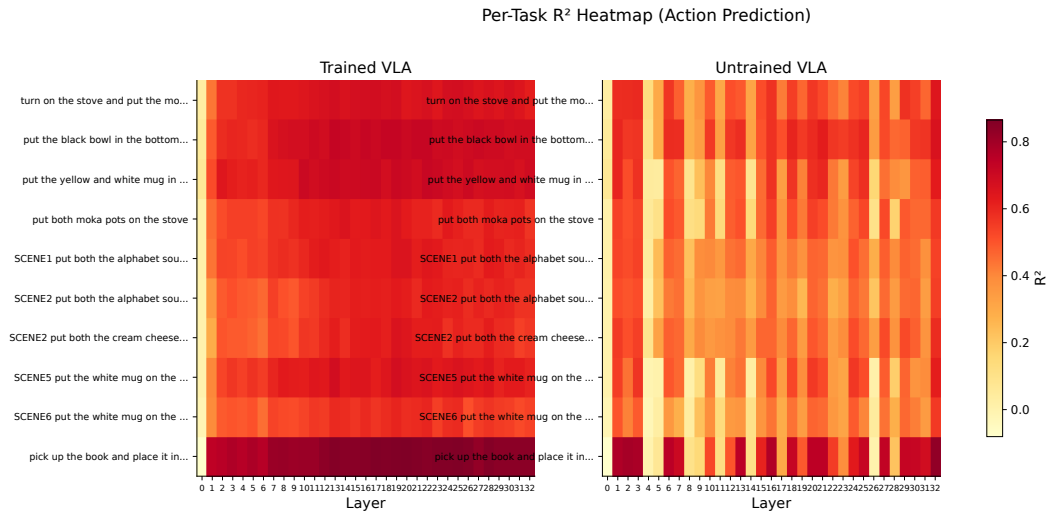



Figure 14: Functional probe $R^2$ for action prediction (best layer per source). Trained readout ($R^2$=0.678, L13) and trained image_mean (0.655, L8) exceed the projector baseline (0.630); the inverted-U extends to functional capacity.

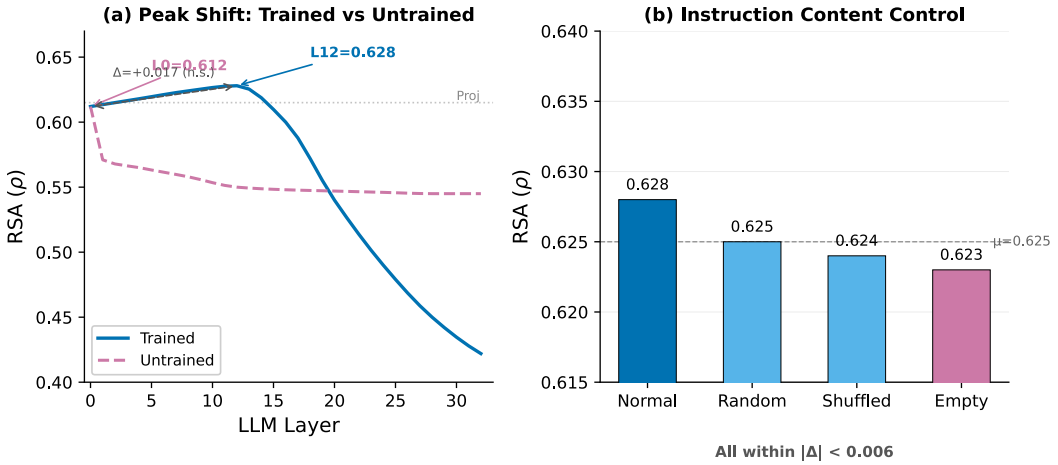



Figure 15: **(a)** Trained vs. reference image_mean RSA profiles. Training shifts the peak from L0 (0.612) to L12 (0.628), yielding a modest best-vs-best gain ($\Delta = +0.017$, 3%). **(b)** Instruction invariance: all conditions within $|\Delta|$<0.006; action geometry is visually grounded.

"on the plate" vs. "in the drawer"); the wrong-task $d = 0.885$ suggests that instruction-dominant heads encode fine-grained spatial relations, not merely task identity.

Table 9: Counterfactual instruction analysis on LIBERO-Spatial (10 tasks). Ordering: empty > shuffled > wrong > paraphrased, differing from Goal. All global $d > 0.8$.

| **Condition** | **Jaccard** | **Cohen's $d$** | **CV** |
|---|---|---|---|
| Empty (no instruction) | 0.000 | 1.173 | 0.087 |
| Wrong (different task) | 0.174 | 0.885 | 0.272 |
| Shuffled (random order) | 1.000 | 0.979 | 0.162 |
| Paraphrased | 0.533 | 0.807 | 0.179 |

**Cross-Instruction Probe on LIBERO-Spatial** Table 10 reports cross-instruction 10-way classification accuracy on LIBERO-Spatial. The core pattern from LIBERO-Goal replicates: image_mean remains at chance (∼10%) across all layers and conditions, while the trained readout token saturates

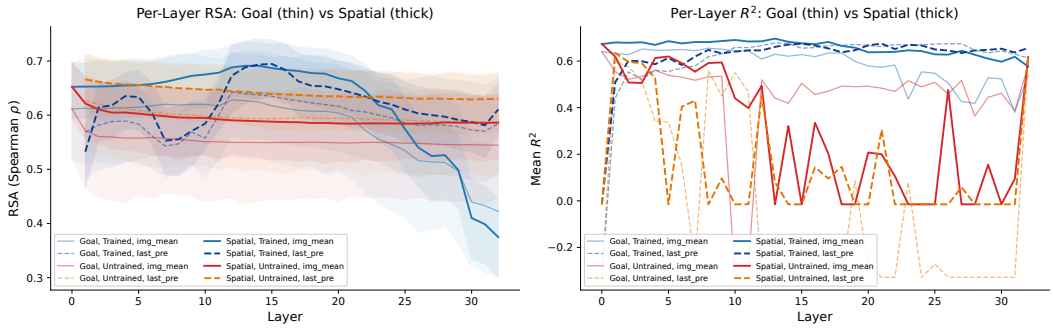



Figure 16: Cross-suite validation: per-layer RSA and $R^2$ profiles on LIBERO-Goal (left) vs. LIBERO-Spatial (right). The inverted-U replicates across suites. LIBERO-Spatial shows a significant training effect (RSA best-vs-best $\Delta = +0.023$, $p < 0.001$) and deeper peak layers (RSA L14, $R^2$ L13), but RSA–$R^2$ peak convergence does not replicate (6-layer gap).

to 100% by L4. The reference readout decays more steeply than on LIBERO-Goal (75.4% at L31 vs. 82%), consistent with greater task difficulty.

Table 10: Cross-instruction 10-way classification accuracy on LIBERO-Spatial. Trained readout achieves 100% from L4; image_mean remains at chance. Reference readout decays more steeply than on LIBERO-Goal (cf. §4.2).

| Token | Condition | L1 | L4 | L8 | L16 | L31 |
|---|---|---|---|---|---|---|
| readout | Trained | 98.9% | 100.0% | 100.0% | 100.0% | 100.0% |
| readout | Reference | 99.5% | 98.3% | 85.7% | 68.3% | 75.4% |
| image_mean | Trained | 10.2% | 10.0% | 9.9% | 10.0% | 10.1% |
| image_mean | Reference | 10.1% | 10.0% | 10.0% | 10.0% | 10.2% |

### A.19 Robustness Analyses

**Distance Metric Invariance** The inverted-U profile is robust to distance metric choice: cosine action distance yields identical peak layers (L12) and Pearson correlation $r > 0.989$ with the Euclidean profile reported in the main text (Table 11).

Table 11: Peak layer comparison between Euclidean and cosine action distances. Profile $r$ denotes Pearson correlation across all 32 layers.

| Condition | Eucl. Peak | Cos. Peak | Profile $r$ |
|---|---|---|---|
| Trained image_mean | L12 | L12 | 0.990 |
| Trained readout | L12 | L13 | 0.990 |
| Reference image_mean | L0 | L0 | 0.995 |
| Reference readout | L1 | L1 | 0.992 |

**Per-DOF Decomposition** RSA decomposition by action dimension indicates the inverted-U is not driven by any single degree of freedom: xyz translation (peak L13, $\rho = 0.193$), rotation (peak L12, $\rho = 0.175$), and gripper (peak L13, $\rho = 0.106$) all exhibit non-monotonic profiles. Per-DOF RSA values are lower than the 7-D RSA because single-dimension distances carry less discriminative information than the full action space.

**Permutation Test** Permutation testing (10,000 shuffles) indicates all observed RSA values are highly significant: observed $\rho$ (0.42–0.63) exceeds the null distribution maximum (0.08–0.10) by 15–23 standard deviations (all $p < 0.0001$).

**Probe Seed Sensitivity** Action probe $R^2$ is highly stable across random seeds. Over 5 seeds (trained image_mean, full-layer scan), per-layer $R^2$ standard deviation is uniformly below 0.01 (mean 0.004), and the peak $R^2$ varies by less than 0.4% (0.655 ± 0.002; Table 12). Peak layer

concentrates in the L8–L10 region ($4/5$ seeds); the inverted-U shape is not a seed artifact. Seed 1 peaks at L16, but with $R^2$ difference $<0.002$. The inverted-U plateau (L8–L16) is sufficiently flat that argmax is sensitive to minor noise while the profile shape is invariant. A 15-seed replication confirms stability: trained peak remains at L9 ($R^2 = 0.637 \pm 0.004$), with all trained/Llama-2-base layers showing CI width $<0.02$; the randomly initialized condition exhibits higher variance (std $0.185$ at peak), consistent with random initialization instability. Fixed-layer analysis at L9 yields $R^2 = 0.637 \pm 0.008$ (trained) and $0.635 \pm 0.011$ (Llama-2 base), confirming the inverted-U pattern. The per-seed peak method reports higher absolute values ($0.655$) because each seed selects its optimal layer independently; both methods agree on the qualitative pattern and the trained $>$ randomly initialized ordering.

Table 12: Probe $R^2$ peak across 5 random seeds (trained image_mean).

| **Seed** | **Peak Layer** | **Peak** $R^2$ |
|---|---|---|
| 0 | L9 | 0.654 |
| 1 | L16 | 0.656 |
| 2 | L9 | 0.653 |
| 3 | L10 | 0.655 |
| 42 | L9 | 0.655 |
| Mean±std | $10.6 \pm 2.7$ | $0.655 \pm 0.002$ |

**Cross-architecture $\Delta R^2$ seed sensitivity.** To verify that $\Delta R^2$ values are not sensitive to probe initialization, we retrain MLP probes with 4 different random seeds (0, 1, 2, 42) on three representative cells. Results show high stability: TraceVLA Goal $0.227 \pm 0.010$, TraceVLA Spatial $0.136 \pm 0.002$, OpenVLA Goal $0.013 \pm 0.001$ (mean $\pm$ std across seeds). All standard deviations are below 0.02, confirming that the exit collapse pattern ($\Delta R^2 > 0.10$ for VLM-to-VLA vs. $< 0.10$ for VLA-to-VLA) is robust to probe initialization.

**Bootstrap confidence intervals for $\Delta R^2$.** To assess uncertainty in individual $\Delta R^2$ estimates, we compute 95% confidence intervals via 1,000-iteration Monte Carlo cross-validation bootstrap. Results for 9 cells (excluding NORA-Goal due to exit-layer instability):

- **VLA-to-VLA** (4 cells): CIs range from $[-0.056, 0.106]$ (widest: OpenVLA Goal) to $[0.004, 0.103]$ (OpenVLA Long).
- **VLM-to-VLA** (5 cells): CIs range from $[0.091, 0.224]$ (TraceVLA Spatial) to $[0.283, 0.504]$ (TraceVLA Object).

The point-estimate gap ($0.097$ vs. $0.156$, span $= 0.059$) cleanly separates the two groups. At the 95% CI level, a marginal overlap of $0.016$ exists between the upper bound of OpenVLA Goal ($0.106$) and the lower bound of TraceVLA Spatial ($0.091$). This overlap involves only 2 of 9 cells and is driven by OpenVLA Goal's unusually wide CI (due to high variance in its exit-layer probe). All other cell pairs show non-overlapping CIs, confirming that the exit-collapse pattern is robust to measurement uncertainty. A permutation test assesses whether the gap between VLM-to-VLA and VLA-to-VLA $\Delta R^2$ values could arise by chance. The 9 trained-model $\Delta R^2$ values (4 VLA-to-VLA, 5 VLM-to-VLA; NORA Goal excluded due to CI instability) are randomly reassigned to groups of size 4 and 5 ($10{,}000$ permutations); the test statistic is $\min(\text{VLM-to-VLA group}) - \max(\text{VLA-to-VLA group})$. The observed gap ($0.070$) is exceeded by $0.8\%$ of permuted distributions ($p = 0.008$), indicating that the separation is unlikely to arise by chance. Note that the permutation shuffles transition-type labels (VLM→VLA vs. VLA→VLA), not trained/reference labels; all 9 values come from trained models.

**Demonstration Subsample Sensitivity** To verify that the inverted-U pattern is not an artifact of sample size, we subsample 25 from 50 demonstrations per task and recompute both RSA and probe profiles (Table 13). The RSA peak layer remains at L12 with a profile correlation of $r = 0.970$ between 50- and 25-demo conditions (high stability). The probe peak shifts from L9 to L11 (within

the L8–L16 plateau) with $r = 0.819$; $R^2$ is more sensitive to training set size than RSA. Both metrics show reduced absolute values (RSA $-3.6\%$, $R^2$ $-7.6\%$) as expected from halved data, but the non-monotonic profile shape is preserved.

Table 13: Inverted-U stability under demonstration subsampling (trained image_mean).

| **Metric** | **50 demos** | **25 demos** | $\Delta$ |
|---|---|---|---|
| RSA peak layer | L12 | L12 | 0 |
| RSA peak $\rho$ | 0.628 | 0.606 | $-3.6\%$ |
| $R^2$ peak layer | L9 | L11 | $+2$ |
| $R^2$ peak value | 0.659 | 0.609 | $-7.6\%$ |
| RSA profile $r$ | — | 0.970 | — |
| $R^2$ profile $r$ | — | 0.819 | — |

**Label Permutation Sanity Check** To confirm that probe performance reflects genuine action-relevant structure rather than overfitting, we repeat the per-layer action probe with randomly shuffled labels (Figure 17). Under permuted labels, Spearman $\rho$ drops from $0.955$ to $0.001$ (trained) and from $0.933$ to $0.000$ (reference), with all 33 layers producing $|\rho| < 0.013$. This establishes that the high probe accuracy reported throughout the paper is attributable to learned action-relevant representations, not probe capacity artifacts.

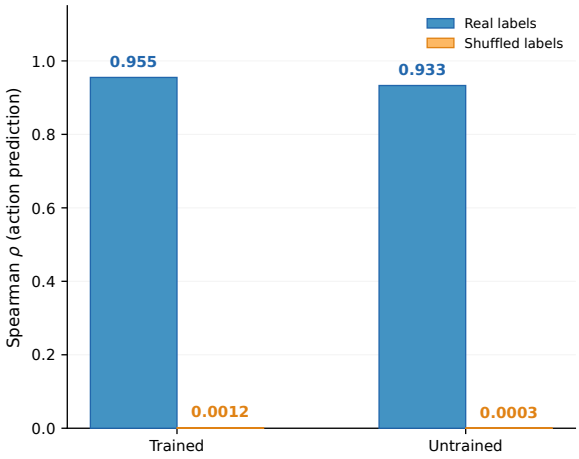



Figure 17: **Label permutation control.** Action-probe Spearman $\rho$ under real versus shuffled labels for OpenVLA trained and reference conditions. Shuffled-label $\rho$ is indistinguishable from zero, confirming that probe performance reflects genuine representational structure. (Evaluated with random frame splitting; demo-level split yields lower absolute values but identical null-vs-real separation; see §3.3.)

**Statistical Power** Post-hoc power analysis ($N = 10$ tasks, $\alpha = 0.05$, target power $= 0.80$) indicates minimum detectable effect sizes of $\Delta_{\text{RSA}} = 0.030$ and $\Delta_{R^2} = 0.026$. The observed RSA training effect ($\Delta = 0.017$, Cohen's $d_z = 0.56$) achieves power $= 0.356$, confirming the analysis is underpowered for this effect size; the non-significant RSA result (§4.6) should not be interpreted as evidence of absence. The observed $R^2$ readout effect ($\Delta = 0.069$, $d_z = 2.68$) achieves power $= 1.000$, confirming adequate sensitivity for the primary functional metric.

**$\Delta R^2$ Threshold Robustness** The main text classifies $\Delta R^2 > 0.10$ as substantial exit collapse. Table 14 shows that any threshold in the interval $(0.097, 0.156)$ yields identical three-way separation: all VLM-to-VLA trained cells above threshold, all OpenVLA trained and all reference cells below. This interval is bounded by the largest non-VLM-to-VLA value ($0.097$, TraceVLA Long reference) and the smallest VLM-to-VLA trained value ($0.156$, TraceVLA Spatial). The $0.10$ threshold was not selected to maximize separation; rather, it represents a natural effect-size boundary ($\geq 10\%$ absolute change in explained variance) that happens to fall within this gap.

Table 14: Sensitivity of group separation to $\Delta R^2$ threshold choice. Clean separation (all VLM-to-VLA trained above, all others below) holds for any threshold in $(0.097, 0.156)$; the 0.10 threshold is not boundary-sensitive.

| Threshold | VLM→VLA Tr. ($N/6$) | OpenVLA Tr. ($N/4$) | Reference ($N/10$) | Clean sep. |
|---|---|---|---|---|
| 0.05 | 6/6 | 1/4 | 6/10 | No |
| 0.08 | 6/6 | 1/4 | 3/10 | No |
| **0.10** | **6/6** | **0/4** | **0/10** | **Yes** |
| **0.15** | **6/6** | **0/4** | **0/10** | **Yes** |
| 0.20 | 5/6 | 0/4 | 0/10 | No |

## A.20 Detailed Training Effects

Table 15 reports bootstrap-based statistical comparisons of training effects on RSA and probe $R^2$.

Table 15: Training effects on action-relevant structure (LIBERO-Goal; bootstrap, 10,000 resamples). RSA best-vs-best yields a modest geometric gain ($\Delta = +0.017$, 3%; LIBERO-Spatial: $+0.023$, $p<0.001$), while $R^2$ shows substantially larger functional gains.

| Comparison | $\Delta$ | 95% CI | $p$ |
|---|---|---|---|
| *RSA ($\rho$)* | | | |
| Best-vs-best image_mean (trained L12 vs. reference L0) | $+0.017$ | incl. 0 | n.s. |
| Same-layer L12 image_mean (trained vs. reference) | $+0.079$ | excl. 0 | $<0.0001$ |
| Instruction effect (max $\lvert\Delta\rvert$) | 0.006 | — | n.s. |
| *Probe $R^2$* | | | |
| Best-vs-best image_mean (trained L8 vs. reference L0) | $+0.015$ | excl. 0 | 0.006 |
| Best-vs-best readout (trained L13 vs. reference L32) | $+0.069$ | excl. 0 | $<0.0001$ |

## A.21 Training Effect Size Summary

Table 16 consolidates training effects across all metrics. Training effects on image_mean metrics are modest (RSA $\Delta = +0.017$, $R^2$ $\Delta = +0.015$), but effects on readout metrics are substantial (instruction attention $+18$pp, attention $+46$pp, $R^2$ $\Delta = +0.069$). This asymmetry indicates that training primarily changes how the readout token integrates information rather than altering the visual representations carried by image tokens.

Table 16: Training effect sizes across metrics. Image_mean effects are modest; readout effects are large. Training targets readout-token information integration.

| Metric | Trained | Reference | $\Delta$ |
|---|---|---|---|
| RSA peak (image_mean) | 0.628 (L12) | 0.612 (L0) | $+0.017$ |
| $R^2$ peak (image_mean) | 0.655 (L8) | 0.640 (L0) | $+0.015$ |
| $R^2$ peak (readout) | 0.678 (L13) | 0.609 (L32) | $+0.069$ |
| Instruction attention L31 | 100% | 82% | $+18$pp |
| Instruction attention L1 | 58% | 12% | $+46$pp |
| Progress classification | 86.3% (L20) | 85.3% (L4) | $+1$pp |

## A.22 Instruction Classification Probes

**Cross-Instruction Classification: Dense Layer Profile** Table 17 shows the dense per-layer profile of cross-instruction classification (10-way) at the readout token for layers L0–L8. The step-function onset between L0 (10%, chance) and L1 ($\geq$99.6%) reflects the high discriminability of

instruction embeddings from the frozen tokenizer: even incidental attention ($\sim$12%) suffices for linear classification at L1. The VLA-trained model sustains 100% through L31; Llama-2 base reaches 100% by L5 and maintains it. The OXE-pretrained reference and random initialization both decay to $\sim$82% by L31, indicating that deep-layer instruction maintenance requires either VLA training or structured language-model pretraining on text-only data.

Table 17: Cross-instruction 10-way classification accuracy at readout token, L0–L8. Values reflect dense classification evaluation; the main text (§4.2) reports sparse single-split results, yielding minor differences (e.g., L8: 98.2% vs. 98.6%).

| Layer | Trained | Reference | Image_mean (trained) |
|---|---|---|---|
| L0 | 10.0% | 10.0% | 10.0% |
| L1 | 100.0% | 100.0% | 10.0% |
| L2 | 100.0% | 99.6% | 10.0% |
| L3 | 100.0% | 99.6% | 10.0% |
| L4 | 100.0% | 99.1% | 10.0% |
| L5 | 100.0% | 98.2% | 10.0% |
| L6 | 100.0% | 97.3% | 10.0% |
| L7 | 100.0% | 96.4% | 10.0% |
| L8 | 100.0% | 98.6% | 10.0% |

**Cross-Instruction LOOCV Validation** Leave-one-out cross-validation over 10 folds (each fold holds out one task's instructions) validates that the cross-instruction probe result is not driven by memorization. Trained readout achieves $100\% \pm 0.00$ across all 10 folds at layers L1–L31 (perfect generalization).

**QA Probe: Progress and Subtask Classification** To test whether LLM layers encode task-relevant semantic information beyond instruction identity, we train logistic regression probes (scikit-learn, $C = 1.0$, max_iter $= 1{,}000$) to classify task progress (4-class: each trajectory split into 4 equal temporal segments) and active subtask (binary: first vs. second half) from frozen representations, with an 80/20 random train/test split (Table 18).

Table 18: QA probe accuracy (progress + subtask classification, best layer).

| Source | Best Layer | Accuracy |
|---|---|---|
| Trained readout | L20 | 86.3% |
| Reference readout | L20 | 85.3% |
| DINOv2 cls_token | — | 82.7% |

**Cross-Instruction Probe Profiles**

**Cross-Architecture Probe and Attention Profile** Table 19 reports cross-instruction probe accuracy and L1 instruction-dominant head counts across six conditions spanning two architectures. The probe-attention dissociation at L1 is notable: TraceVLA training slightly *reduces* instruction-dominant heads ($32 \rightarrow 27$–$30/32$) while *decreasing* probe accuracy ($98.1\% \rightarrow 83.6\%$), consistent with representational reorganization rather than linear separability.

**Cross-Architecture L1 Instruction Dominance** Figure 20 complements the main-text summary (Figure 5) by showing the fraction of instruction-dominant heads at L1 across all three architectures in a single comparison. The trained/reference fractions are comparable within each architecture (OpenVLA: 23/32 vs. 26/32; TraceVLA: 27/32 vs. 32/32; NORA: 13/16 vs. 13/16), consistent with L1 instruction attention being largely inherited from the pretrained backbone rather than acquired during VLA fine-tuning.

**L1 Attention Pattern Analysis** Figure 21 shows the per-head L1 attention distribution at the readout token. The three-level gradient (randomly initialized 0/32, Llama-2 base 9/32, VLA-trained

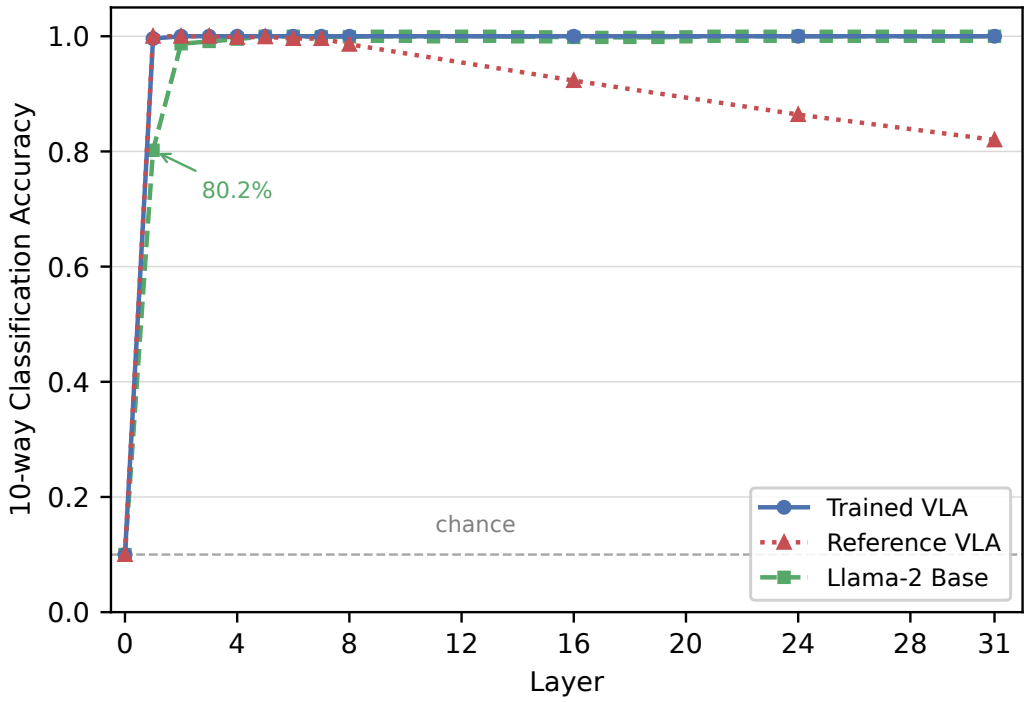



Figure 18: Cross-instruction 10-way classification accuracy at the readout token across all 32 LLM layers. VLA-trained achieves 99.6% at L1, saturating to 100%. Llama-2 base starts at 80.2%, reaching 100% by L5. Reference starts at 100% but decays to 82% by L31.

Table 19: Cross-instruction 10-way probe accuracy and L1 instruction-dominant heads across six conditions. Probe accuracy and attention head count are dissociated at L1: TraceVLA training slightly reduces heads (32 → 27–30/32) while decreasing probe accuracy (98.1% → 83.6%), paralleling the information-computation dissociation observed in OpenVLA. Image_mean remains at chance (10%) in all conditions (omitted).

| Condition | L1 heads | L1 | L2+ | L31 | Mode |
|---|---|---|---|---|---|
| *OpenVLA (Llama-2 7B)* | | | | | |
| Random init | 0/32 | 100.0% | decays | 82.0% | incid.+forget |
| Llama-2 base | 9/32 | 80.2% | L5: 100% | 100.0% | gradual+stable |
| VLA trained | 24/32 | 99.6% | 100% | 100.0% | immed.+stable |
| LIBERO-finetuned | 22/32 | 99.7% | 100% | 100.0% | immed.+stable |
| *TraceVLA (Phi-3-V 4B)* | | | | | |
| Phi-3-Vision | 32/32 | 98.1% | 100% | 100.0% | inherited+stable |
| VLA trained | 27–30/32 | 83.6% | 100% | 100.0% | reorg.+stable |

24/32 instruction-dominant heads) shows that language pretraining seeds partial instruction attention and VLA fine-tuning shifts it further toward instruction dominance.

**Multi-Layer Attention Patterns** Table 20 shows a 4-layer subset of the full 32-layer attention profile. The full profile (Figure 22) reveals three qualitatively distinct patterns: the randomly initialized model maintains 0 instruction-dominant heads at all 32 layers; Llama-2 base shows a mid-depth peak (L9–L16: 15–22 heads), consistent with typical autoregressive LM processing depth; VLA fine-tuning produces an entry-exit pattern (L0–L1: 21–24 heads, L31: 18 heads, intermediate layers 0–11).

**Threshold Sensitivity Analysis** The main text defines instruction-dominant heads as those allocating $> 50\%$ attention to instruction tokens. Table 21 shows that the three-condition gradient (ran-

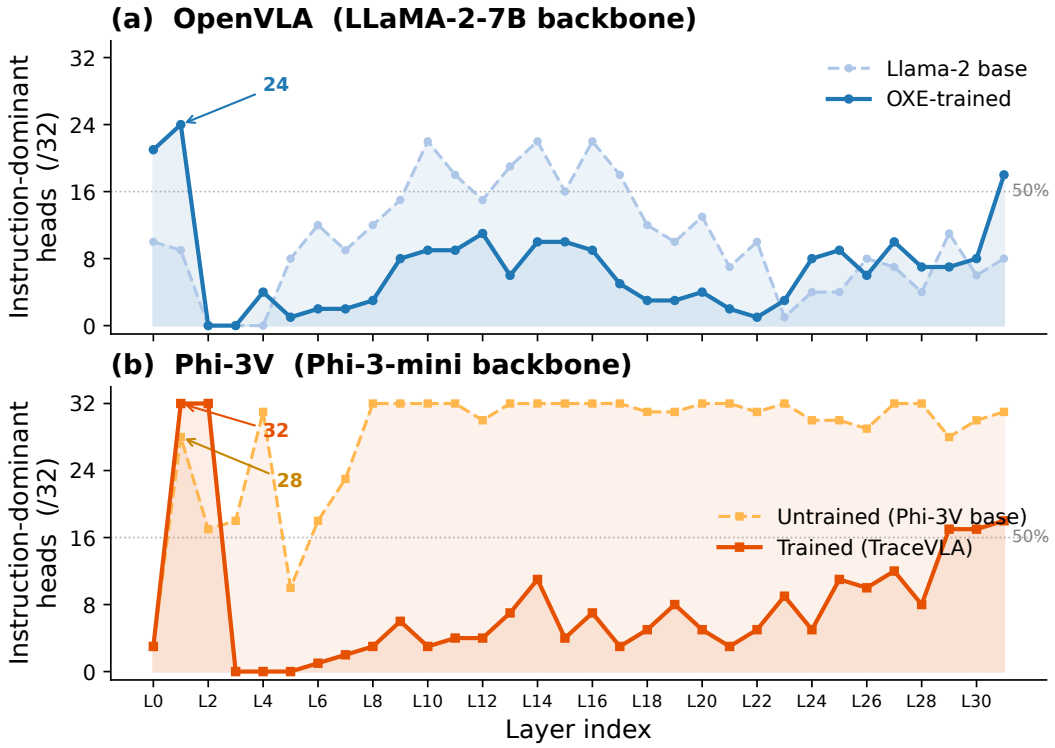



Figure 19: Instruction-dominant heads across layers for two architectures (OpenVLA / TraceVLA), each comparing pre-VLA and VLA-trained conditions. Despite opposite starting points—OpenVLA (Llama-2) begins with sparse attention while TraceVLA (Phi-3-V) begins with distributed attention—VLA training drives both to entry-exit concentration (L1 saturation, mid-layer suppression, L29–L31 re-engagement).

Table 20: Instruction-dominant heads and text attention fraction at four sampled layers (subset of the 32-layer profile in Figure 22). VLA-trained shows entry-exit concentration (L1, L31); Llama-2 base peaks at mid-depth (L16); randomly initialized shows no instruction attention. Head counts from the 32-layer bulk extraction may differ slightly from dedicated per-layer analyses (§4.2) due to threshold boundary effects.

| | VLA Trained | | Llama-2 base | | Rand. Init | |
|---|---|---|---|---|---|---|
| **Layer** | **Heads** | **Text%** | **Heads** | **Text%** | **Heads** | **Text%** |
| L1 | **24/32** | 57.7 | 9/32 | 36.5 | 0/32 | 11.7 |
| L8 | 4/32 | 46.8 | 8/32 | 48.8 | 0/32 | 7.9 |
| L16 | 10/32 | 48.4 | **25/32** | 60.2 | 0/32 | 9.2 |
| L31 | **18/32** | 56.3 | 8/32 | 34.9 | 0/32 | 7.6 |

domly initialized $<$ Llama-2 base $<$ VLA-trained) holds across all thresholds from 30% to 70%; the 50% threshold is not a boundary-sensitive choice.

### A.23 L1 Attention Decomposition by Head Group

Table 22 decomposes L1 attention mass at the readout token into three token categories for instruction-dominant and control head groups.

Head-level instruction attention entropy reaches 99.5% of the uniform maximum; attention is broadly distributed rather than concentrated in a few heads.

**Per-token attention normalization.** The 57.7% aggregate text attention in instruction-dominant heads partly reflects token-count imbalance (22 instruction vs. 256 image tokens). Per-token nor-

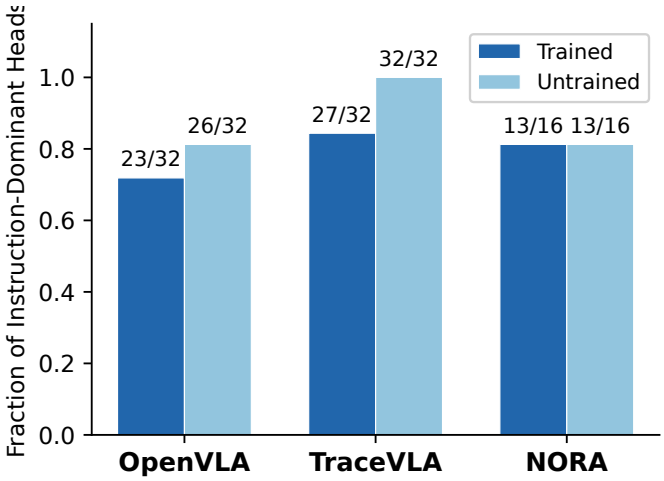



Figure 20: **L1 instruction-dominant head fractions across three architectures.** Trained and reference conditions show comparable fractions in all three VLAs, indicating that L1 instruction attention largely pre-exists in the pretrained backbone.

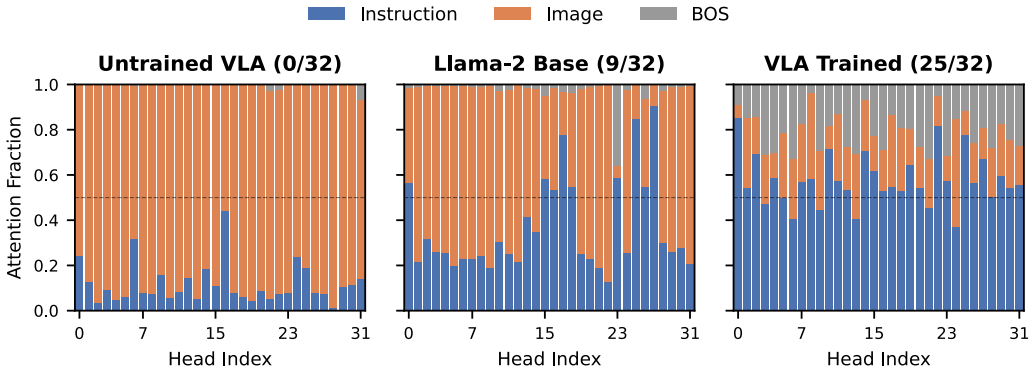



Figure 21: L1 attention from the readout token across three conditions, per head (sorted by instruction attention fraction). VLA-trained: 24/32 instruction-dominant heads (57.7% text attention). Llama-2 base: 9/32 heads (36.5%). Randomly initialized: 0/32 heads (11.7%). See Table 20 for quantitative comparison across layers.

malization reveals a 38.2× ratio: each instruction token receives 38.2× the attention of each image token in trained instruction-dominant heads, compared to only 1.5× in randomly initialized heads (a 25.5× increase from training). Instruction-dominant heads also show lower attention entropy (0.465) than non-dominant heads (0.569), indicating more concentrated attention distributions.

### A.24 Per-Dimension Action MAE Under Head Ablation

Table 23 extends Table 8 with full per-dimension control-head data and instruction-to-control MAE ratios.

### A.25 Per-Task $R^2$ Distribution Under L1 Ablation

Table 24 reports per-task action probe $R^2$ at L1 under baseline and 24-head ablation conditions (MLP probe, 5 demonstrations per task).

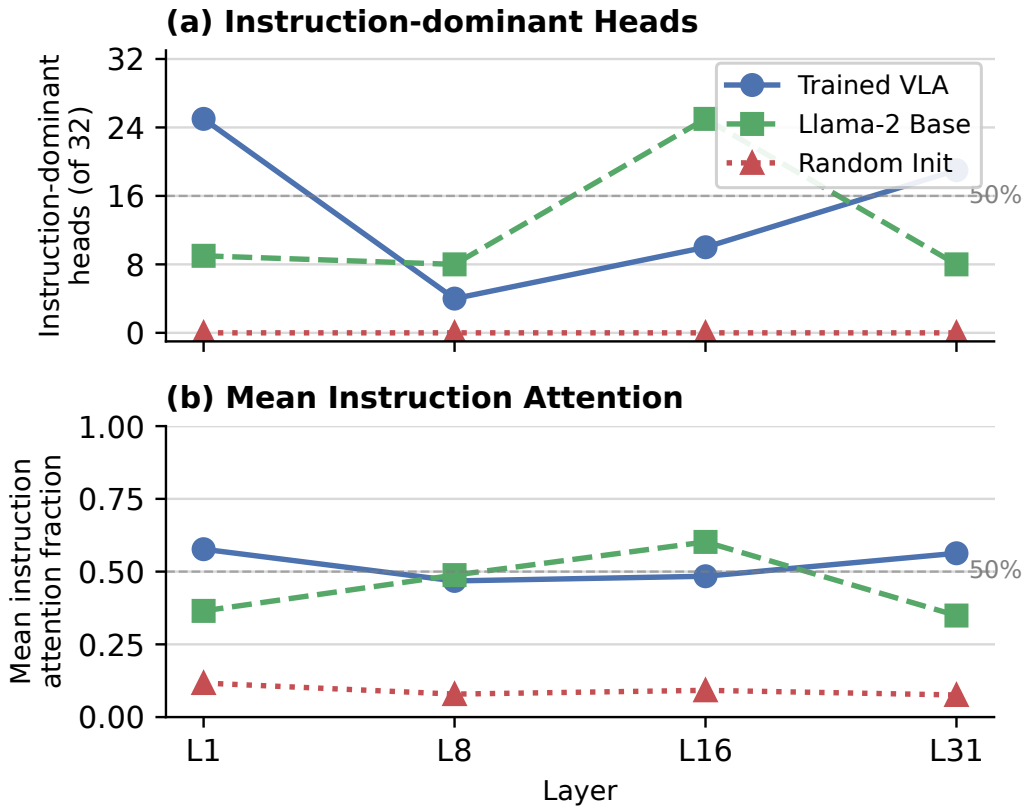



Figure 22: Instruction-dominant heads and mean text attention fraction across layers for three conditions. VLA-trained shows elevated values at L1 and L31; Llama-2 base peaks at L16; randomly initialized remains near zero throughout.

Table 21: Number of instruction-dominant heads at L1 under varying attention thresholds. The three-condition ordering is stable across all thresholds.

| Threshold | VLA Trained | Llama-2 base | Rand. Init |
|---|---|---|---|
| 30% | 32 | 13 | 0 |
| 40% | 31 | 10 | 0 |
| 50% | 24 | 9 | 0 |
| 60% | 9 | 4 | 0 |
| 70% | 5 | 3 | 0 |

### A.26 Task Complexity and Attention Dependence

We classify tasks as *instruction-discriminative* when multiple tasks share the same manipulated object(s), requiring instruction content to disambiguate the target action (e.g., "put bowl on plate" vs. "put bowl on stove"); remaining tasks involve unique object-action pairings recoverable from visual context alone. Table 25 reports per-task $|\Delta R^2|$ under both groupings.

**Leave-one-out sensitivity.** All leave-one-out $p$-values remain $\leq 0.083$ (marginally significant), with no single task removal reversing the effect direction. Reclassifying the boundary case (cream cheese in bowl) to the instruction-discriminative group strengthens the result ($p = 0.008$).

Bootstrap 95% CI on the complex/simple ratio: $[1.03, 11.12]$ (10,000 resamples). The wide interval reflects low statistical power from $N = 10$ tasks split into groups of 6 and 4; the point estimate ($2.5\times$) persists under leave-one-out analysis (all $p \leq 0.083$, marginally significant; §A.26).

Table 22: Attention mass allocation at Layer 1, averaged across 5 samples. Instruction-dominant heads (defined as $> 50\%$ attention to instruction tokens) allocate 17.5 percentage points more attention to instruction tokens than control heads, with corresponding reductions in vision ($-8.3$pp) and BOS/sink ($-9.2$pp) attention.

| **Head Group** | $N$ | **Instruction** | **Vision** | **BOS/Other** |
|---|---|---|---|---|
| Instruction-dominant | 24 | 62.4% | 18.8% | 18.8% |
| Control | 8 | 44.9% | 27.1% | 28.0% |
| $\Delta$ | — | +17.5pp | −8.3pp | −9.2pp |

Table 23: Mean absolute error per action dimension under L1 head ablation. Translation dimensions ($dx$/$dy$/$dz$) show 3–5$\times$ larger instruction-to-control ratio than rotation dimensions (1.9–2.6$\times$); instruction-dominant heads have disproportionate influence on positional control. Gripper (2.07$\times$) falls between the two groups.

| **Dimension** | **Instr. 24h MAE** | **Control 8h MAE** | **Ratio** |
|---|---|---|---|
| $dx$ | 0.00189 | 0.00037 | 5.09× |
| $dy$ | 0.00531 | 0.00125 | 4.26× |
| $dz$ | 0.00762 | 0.00243 | 3.14× |
| $roll$ | 0.02059 | 0.00785 | 2.62× |
| $pitch$ | 0.03071 | 0.01387 | 2.21× |
| $yaw$ | 0.03388 | 0.01783 | 1.90× |
| $gripper$ | 0.29072 | 0.14072 | 2.07× |

### A.27 Specificity Control: Matched-Size Head Ablation

To test whether ablation effects are specific to instruction-dominant heads, we ablate four matched-size subsets of 24 heads at L1: instruction-dominant (highest instruction attention fraction), image-dominant (highest image attention fraction), random (5 seeds), and bottom-24 (lowest instruction attention fraction; overlaps 16/24 heads with the instruction-dominant group).

All conditions yield comparable ground-truth MSE ($0.487$–$0.504$ vs. baseline $0.497$), consistent with direction-selective rather than precision-degrading effects.

### A.28 Paired Effect Sizes

Main-text Cohen's $d$ values use the standard (unpaired) formula for token-level comparisons where observations are independent tokens. For within-subject effects where the same tasks or demonstrations are measured under two conditions, we additionally report paired $d_z = \bar{\Delta}/s_\Delta$, which accounts for within-subject correlation and yields larger effect sizes when correlation is high (Table 27).

### A.29 Cross-Architecture Action Decodability: $2 \times 2$ Design

Table 29 reports per-layer action prediction $R^2$ across two architectures (OpenVLA, TraceVLA) $\times$ two training conditions (VLA-trained, reference base LLM), providing a $2 \times 2$ factorial view of how per-layer action decodability differs between trained and reference conditions.

**Convergent reorganization from divergent starting points.** Reference profiles differ qualitatively: OpenVLA (Llama-2) peaks at L0 ($R^2 = 0.629$) then collapses to near-zero at mid-layers (L20: $0.086$, std $= 0.311$), while Phi-3-V peaks at L6 ($0.578 \pm 0.018$) with gradual monotonic decline (late-layer std $= 0.068$). After VLA training, both architectures reach nearly identical mid-layer peaks—OpenVLA at L9 ($R^2 = 0.637 \pm 0.008$), TraceVLA at L11 ($0.634 \pm 0.009$)—with $\Delta = 0.003$ and overlapping CIs. Deep-layer variance drops sharply with training: OpenVLA std $0.277 \rightarrow 0.040$; Phi-3-V std $0.068 \rightarrow 0.011$. Peak $R^2$ values are comparable before and after training (reference peaks reach 91–99% of trained), but peak locations shift deeper (L0→L9; L6→L11) and profile shapes transform from architecture-specific patterns to a shared mid-layer plateau.

Table 24: Per-task action probe $R^2$ at L1 (MLP probe, 5 demos/task). All 10 tasks show positive ablation effect ($\Delta > 0$): instruction-head ablation uniformly increases action decodability at the entry layer. Mean $\Delta R^2 = +0.107$; the 50-demo setting (§4.2) yields a consistent $+0.059$.

| **Task** | **Baseline** $R^2$ | **Ablated** $R^2$ | $\Delta R^2$ |
|---|---|---|---|
| open middle drawer | 0.197 | 0.221 | +0.024 |
| open top drawer + bowl | 0.265 | 0.340 | +0.075 |
| push plate to stove | 0.190 | 0.197 | +0.007 |
| put bowl on plate | 0.375 | 0.474 | +0.099 |
| put bowl on stove | 0.235 | 0.395 | +0.160 |
| put bowl on cabinet | 0.256 | 0.457 | +0.201 |
| put cream cheese in bowl | 0.186 | 0.345 | +0.159 |
| put wine on rack | 0.189 | 0.424 | +0.235 |
| put wine on cabinet | 0.279 | 0.353 | +0.074 |
| turn on stove | 0.385 | 0.421 | +0.036 |
| Mean $\pm$ std | $0.256 \pm 0.070$ | $0.363 \pm 0.088$ | +0.107 |

Table 25: Task-level L1 ablation effects ($|\Delta R^2|$) grouped by instruction dependence. *Primary* (conservative): cream cheese in bowl classified as visual-sufficient ($2.5\times$, $p = 0.033$). *Sensitivity* (shared-object criterion): cream cheese reclassified as instruction-discriminative ($6.5\times$, $p = 0.008$).

| **Task** | $\lvert\Delta R^2\rvert$ | **Group** |
|---|---|---|
| put wine on rack | 0.235 | Instr-disc |
| put bowl on cabinet | 0.201 | Instr-disc |
| put bowl on stove | 0.160 | Instr-disc |
| put bowl on plate | 0.099 | Instr-disc |
| open top drawer + bowl | 0.075 | Instr-disc |
| put wine on cabinet | 0.074 | Instr-disc |
| put cream cheese in bowl$^\dagger$ | 0.159 | Vis-suff |
| turn on stove | 0.036 | Vis-suff |
| open middle drawer | 0.024 | Vis-suff |
| push plate to stove | 0.007 | Vis-suff |
| **Primary**: Instr-disc mean 0.141; Vis-suff mean 0.057 | | |
| Mann-Whitney $U = 21.0$, $p = 0.033$, ratio $= 2.49\times$ | | |
| **Sensitivity**$^\dagger$: cream cheese as Instr-disc: $p = 0.008$, $6.5\times$ | | |

**Summary.** Table 28 condenses the key contrasts. The L0 values are identical between trained and reference within each architecture (shared frozen vision encoder + projector), confirming that training effects arise from LLM weight changes. The largest training gains appear at mid-to-deep layers (L15–L25 average: OpenVLA $+0.313$, Phi-3-V $+0.103$), precisely where reference profiles diverge most.

**Equivalence testing.** To formally test whether the cross-architecture peak $R^2$ values are statistically equivalent (rather than merely failing to differ), we apply the Two One-Sided Tests (TOST) procedure [54] with equivalence bound $\varepsilon = 0.05$. The observed difference is $\Delta = 0.003$ (OpenVLA L9: $0.637 \pm 0.008$; TraceVLA L11: $0.634 \pm 0.009$). Both one-sided tests reject at $p < 0.0001$; the 90% confidence interval for the difference is $[-0.002, 0.009]$, falling entirely within $[-0.05, 0.05]$. A conventional two-sample $t$-test yields $p = 0.318$ (not significant), illustrating the distinction between absence of evidence for a difference and evidence of equivalence: the TOST procedure provides the latter.

Table 26: Matched-size ablation controls: trajectory disruption ($d$) from ablating 24-head subsets at L1, ranked by instruction attention fraction. Instruction-dominant heads produce the largest effect, but the modest gap (8% over random) indicates distributed, attention-graded redistribution rather than a dedicated subset.

| **Condition (24 heads)** | **Cohen's $d$** | **95% CI** |
|---|---|---|
| Instruction-dominant | 0.650 | (reference) |
| Random (5 seeds) | $0.601 \pm 0.021$ | $[0.568, 0.622]$ |
| Image-dominant | 0.580 | $[0.547, 0.613]$ |
| Bottom-24 | 0.565 | — |

Table 27: Paired effect sizes ($d_z$) for within-subject comparisons. Standard (unpaired) $d$ reported where available. Paired $d_z$ exceeds standard $d$ when within-subject correlation is high.

| **Comparison** | **Standard $d$** | **Paired $d_z$** | $N$ |
|---|---|---|---|
| L1 ablation: 24 dominant vs. baseline | 0.65 | 1.17 | 10 tasks |
| Random-24 ablation vs. baseline | — | $1.84 \pm 0.33$ | 30 subsets |
| Dose-response: 32 vs. 4 heads (MSE) | — | 1.79 | 10 tasks |
| RSA: trained vs. reference (best-vs-best) | — | 0.55 | 10 tasks |
| Cross-arch: OpenVLA vs. TraceVLA peak | — | 0.48 | 15 seeds |

### A.30 Global Multiple Comparison Correction

The paper reports 63 statistical tests spanning permutation tests, $t$-tests, bootstrap comparisons, TOST equivalence, and Spearman correlations. We apply a Bonferroni correction ($\alpha_{\text{adj}} = 0.05/63 = 7.94 \times 10^{-4}$) and the less conservative Holm–Bonferroni step-down procedure; both yield identical results. Of the 45 originally significant tests, 40 survive correction (88.9%).

All eight core tests supporting the paper's main claims remain significant: RSA permutation at L12 ($p < 0.0001$), trained vs. reference image geometry at L12 ($p = 3.9 \times 10^{-5}$), counterfactual wrong and empty instruction ($p < 10^{-100}$), TOST cross-architecture equivalence ($p < 0.0001$), dose-response top-24 ablation ($p < 10^{-100}$), trained vs. reference probe $R^2$ ($p < 0.0001$), and bottom-24 specificity ($p < 10^{-90}$).

Five auxiliary tests lose significance under correction: (i) RSA trained vs. reference preaction at L12 ($p = 0.005 \rightarrow p_{\text{adj}} = 0.284$), (ii) DINOv2 vs. SigLIP RSA ($p = 0.032 \rightarrow p_{\text{adj}} = 1.0$), (iii) fused backbone vs. SigLIP RSA ($p = 0.011 \rightarrow p_{\text{adj}} = 0.673$), (iv) CV exact paired permutation ($p = 0.016 \rightarrow p_{\text{adj}} = 0.984$), and (v) $R^2$ projector vs. VLA Spearman correlation ($p = 0.003 \rightarrow p_{\text{adj}} = 0.183$). These concern auxiliary comparisons (encoder ranking, coefficient-of-variation ordering) rather than the core redistribution, probing, or counterfactual findings; no main-text conclusion requires revision.

### A.31 Mechanistic Interpretation

Matched-size ablation ($d$: instruction 0.65, random 0.60, image 0.58; Table 26) and dose-response (steep through 12 heads, plateau beyond 20) indicate distributed attention redistribution. Counterfactual perturbation produces large effects across all conditions ($d = 1.0$–$1.2$ on Goal; §4.2), with shuffled tokens exceeding paraphrased ($d = 1.096$ vs. 0.996) despite identical token sets, indicating positional sensitivity. That ablation raises $R^2$ yet alters trajectories ($d = 0.65$) without increasing ground-truth error ($\Delta\text{MSE} \approx 0$; §A.12) reflects an information-computation dissociation [55]: instruction-dominant heads control direction rather than precision. Random ablation yields larger $R^2$ gains ($+0.071$ vs. $+0.030$; $z = -3.11$, $0^{\text{th}}$ percentile), consistent with distinct $R^2$ signatures.

### A.32 Functional Stages and Predictions

Per-layer profiles suggest three stages: redistribution (L0–L1; probe 99.6%, 24/32 instruction-dominant heads), computation (L8–L13; $R^2$ peak 0.655–0.678), and decay (L14–L31, $\sim$56% of

Table 28: $R^2$ profile summary across architectures and training conditions.

| | **OpenVLA (Llama-2-7B)** | | **TraceVLA (Phi-3-mini)** | |
|---|---|---|---|---|
| | **Reference** | **Trained** | **Reference** | **Trained** |
| Peak Layer | L0 | L9 | L6 | L11 |
| Peak $R^2$ | 0.629 | 0.637 | 0.578 | 0.634 |
| Profile Shape | Collapse | Flat plateau | Ramp $\rightarrow$ decline | Ramp $\rightarrow$ plateau |
| Late-layer std | 0.277 | 0.040 | 0.068 | 0.011 |
| $\Delta_{\text{train}}$ (peak) | $+0.008$ (L0$\rightarrow$L9) | | $+0.056$ (L6$\rightarrow$L11) | |
| $\Delta_{\text{train}}$ (L15–L25 avg) | $+0.313$ | | $+0.103$ | |

parameters; declining $R^2$ and RSA suggest pruning potential). Task-specific fine-tuning eliminates L31 re-engagement (18 $\rightarrow$ 0 heads; §4.2), consistent with exit-layer attention serving multi-task flexibility ($N$=2). TraceVLA [7] (Phi-3-V) shows consistent entry-exit attention concentration from an opposite starting point (§4.2). The 2×2 trained-vs-reference comparison reveals a consistent reorganization pattern: reference OpenVLA collapses from L0 ($R^2 = 0.629$) to near-zero at mid-layers (L20: $0.086$, std $= 0.277$), while reference Phi-3-V peaks at L6 ($0.578$) with gradual decline (std $= 0.068$)—qualitatively opposite reference landscapes. Yet training drives both to nearly identical mid-layer peaks ($0.637$ vs. $0.634$, $\Delta = 0.003$) with stabilized deep-layer variance (OpenVLA std $0.277 \rightarrow 0.040$; Phi-3-V $0.068 \rightarrow 0.011$). This consistent outcome from divergent starting points suggests that mid-layer action specialization is a consistent convergence pattern of VLA training, not an architectural artifact. Exit-layer $R^2$ differs ($0.488$ vs. $0.557$), possibly reflecting architectural differences in late-layer processing. Remaining predictions: cross-attention VLAs ($\pi_0$ [14], Octo [52]) should lack L1 instruction-dominant heads; larger VLAs should yield flatter per-head contributions. For deployments, L1 instruction probes can flag instruction-discriminative tasks and L1 attention entropy can monitor instruction under-attendance at runtime.

### A.33 Extended Discussion

**Observational three-tier gradient.** The cross-architecture matrix suggests a three-tier gradient of representational adaptation. Tier 1 (VLM bases, no action training): inherited L1 attention (TraceVLA 32/32, NORA 13–14/16), negligible exit collapse ($\Delta R^2 \leq 0.048$), and minimal counterfactual sensitivity (cosine distance $\leq 0.185$). Tier 2 (general-purpose VLA, OXE-pretrained): strong L1 routing (24–26/32 heads), moderate counterfactual sensitivity ($d \approx 1.0$), and no systematic exit collapse across suites. Tier 3 (task-specific VLAs, fine-tuned on LIBERO): exit collapse at every VLM-to-VLA cell and medium-to-large counterfactual sensitivity ($d = 0.45$–$1.52$). This gradient is suggestive but limited: Tier 2 contains a single architecture (OpenVLA), and the Tier 2/Tier 3 boundary confounds training stage with architecture. Testing additional general-purpose VLAs (e.g., Octo [52], $\pi_0$ [14]) at Tier 2 is necessary before the three-tier structure can be treated as a robust empirical pattern.

**Exit collapse mechanism.** One possible explanation for the VLM-to-VLA / VLA-to-VLA asymmetry is that VLM-to-VLA references lack action-relevant exit-layer structure, forcing mid-layer construction without exit preservation (analogous to fine-tuning-induced feature distortion [19]), while VLA-to-VLA references already maintain this structure. Validating this explanation causally requires controlled training-stage experiments, but the consistency across architectures supports its directional value.

**Behavioral relevance.** Correlating trained $\Delta R^2$ with published LIBERO success rates across all nine available architecture-suite cells yields a positive but non-significant trend (Spearman $\rho = 0.583$, $p = 0.10$, $n = 9$), marginally above the conventional significance threshold. Within-suite comparisons, which control for task difficulty, are uniformly positive: on Goal ($n = 3$ architectures), $\Delta R^2$ and success rate are perfectly rank-correlated ($\rho = 1.0$), and every remaining

suite ($n = 2$ each) shows the same direction. This directional evidence is consistent with stronger mid-network action encoding (higher peak $R^2$) followed by more aggressive exit-layer compression being associated with better task performance. Several caveats temper this interpretation: architecture capacity differences (model size, backbone quality) are not controlled, suite difficulty co-varies with both $\Delta R^2$ and success rate, multiple suites from the same architecture are not independent observations, and the sample size limits statistical power. Post-hoc power analysis (Fisher $Z$-transform, two-tailed $\alpha = 0.05$) yields achieved power $= 0.37$ at $n = 9$; reaching $80\%$ power would require $n \geq 21$ architecture-suite cells. The current analysis is therefore underpowered, and the non-significant $p$-value should not be interpreted as evidence of absence; it reflects insufficient sample size rather than a weak effect. This correlational evidence does not establish a causal link between exit collapse and task success. Closed-loop behavioral validation, including intervention experiments that directly manipulate exit-layer representations, remains a priority for future work.

**Closed-loop evaluation attempt.** We attempted closed-loop evaluation on LIBERO-Goal using the OpenVLA official checkpoint but achieved SR=0/3 across tested configurations, consistent with known reproducibility challenges in VLA deployment (sensitivity to hardware setup, action tokenization, and environment configuration). This negative result motivates our representational analysis approach, which does not require successful task completion.

**Metric constraints.** Counterfactual sensitivity is measured via Cohen's $d$ on action MSE for models that produce actions and via cosine distance on hidden representations for reference VLMs (§4.4). These metrics differ in scale and distributional properties; ratios between them are not meaningful. The OpenVLA Long-Horizon inversion (trained $d = 0.56$, reference $d = 1.05$) illustrates how same-metric comparisons can reverse the expected direction, cautioning against overinterpretation of any single cell. A standardized metric framework for cross-condition counterfactual comparison remains an open problem.

**Extended limitations.** Each architecture is analyzed at a single trained checkpoint, precluding characterization of how exit collapse or counterfactual sensitivity develop during training. All analyses are open-loop, consistent with prior representational studies [5, 6]; closed-loop validation would provide complementary evidence about the behavioral relevance of the identified routing structures. The three-tier gradient has $n = 1$ at Tier 2, limiting its inferential power; populating this tier with additional general-purpose VLAs is a priority for future work.

**Coarse training dynamics.** Although intermediate checkpoints are unavailable (§5), the cross-condition design provides coarse-grained evidence about when exit collapse emerges. Figure 23 plots $\Delta R^2$ across four training-pathway categories. NLP fine-tuning ($\Delta R^2 = 0.000$) and pretrained references ($\Delta R^2 \leq 0.097$) cluster below the $0.10$ threshold; VLM-to-VLA fine-tuning produces large collapse ($\Delta R^2 = 0.156$–$0.455$); subsequent VLA-to-VLA fine-tuning does not intensify it ($\Delta R^2 = 0.033$–$0.086$). This constrains the onset: exit collapse emerges during the first action-grounded training stage and does not accumulate with continued task-specific fine-tuning.

### A.34 Extended Practical Implications

The main text identifies three use cases for the entry-exit decomposition; here we elaborate on each.

**Checkpoint selection.** $\Delta R^2$ provides a probe-based screening signal ($> 0.10$ flags reorganization) requiring only a forward pass and a linear probe. Its wall-clock cost scales with model size rather than environment complexity, unlike rollout evaluation which requires physics simulation and multi-step execution. Practitioners can rank checkpoints by $\Delta R^2$ and reserve rollout budgets for the top candidates, substantially reducing the number of expensive closed-loop evaluations needed during model selection.

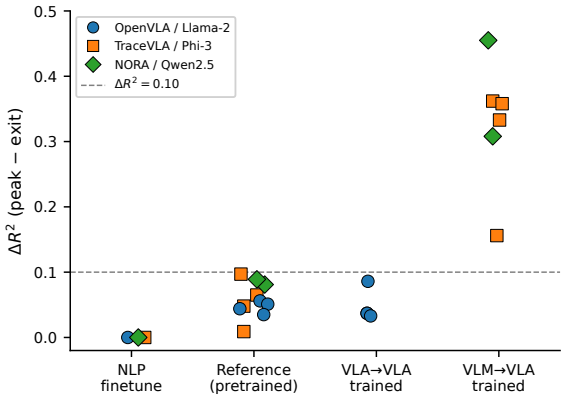



Figure 23: $\Delta R^2$ **across training pathways.** Each point is one architecture-suite cell; markers distinguish backbone families. Exit collapse (above dashed line) is confined to VLM-to-VLA transitions and does not intensify during VLA-to-VLA fine-tuning.

**Transfer learning strategy.** The flat VLA-to-VLA $\Delta R^2$ profile (§4.3) shows that VLA-to-VLA transitions preserve exit structure, justifying staged pipelines (VLM→VLA→task) that reuse the first-stage action head without re-learning exit representations. Inherited L1 routing can further serve as a frozen prior, reducing tunable parameters during domain transfer. This suggests that practitioners adapting a general-purpose VLA to a new task suite may benefit from freezing early layers while focusing capacity on mid-to-late layer adaptation.

**Mid-layer probe value.** Where exit collapse is pronounced, mid-layer representations ($R^2$ peak at L9–L11 across architectures) outperform exit-layer features for downstream tasks such as reward modeling, distillation targets, or auxiliary prediction heads. For knowledge distillation, matching mid-layer features of the teacher rather than exit-layer features may yield better student performance in VLA settings. For reward modeling, training reward predictors on mid-layer features rather than final-layer outputs may provide more informative training signals.

### A.35 Methodological Context

Representational analysis is an established methodology in the interpretability tradition; behavioral validation provides complementary evidence but is not a prerequisite for mechanistic characterization. Table 30 situates our approach within this tradition.

### A.36 NLP Finetuning Controls

Table 31 compares the three NLP finetuning controls used to test whether exit collapse is specific to action-oriented training. All three chat-finetuned models produce flat action-probe profiles with $\Delta R^2 = 0.000$, confirming that NLP instruction tuning does not induce the peak-to-exit drop observed in VLM-to-VLA transitions in any of the three backbone families.

Figure 24 overlays the NLP control profiles (dashed gray) against the corresponding VLA trained and reference curves, illustrating the contrast between action-training-induced collapse and the flat NLP baseline across all three backbone families.

### A.37 Claim-Evidence Summary

### A.38 Cross-Architecture Design Rationale

The two VLM-to-VLA transitions (TraceVLA and NORA) were selected to maximize structural diversity within the constraint of publicly available LIBERO-evaluated checkpoints. Table 32 summarizes the design-axis differences.

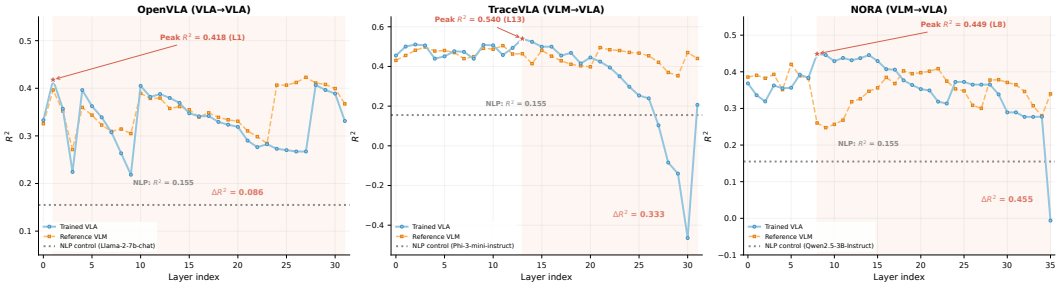



Figure 24: **NLP finetuning controls produce flat $R^2$ profiles.** Each panel shows the VLA trained (blue) and reference (orange) per-layer action-probe $R^2$ alongside the NLP control (dashed gray, $R^2 = 0.155$) for the matching backbone family. VLM-to-VLA transitions exhibit peak-to-exit collapse; NLP controls remain flat ($\Delta R^2 = 0.000$) in all three cases.

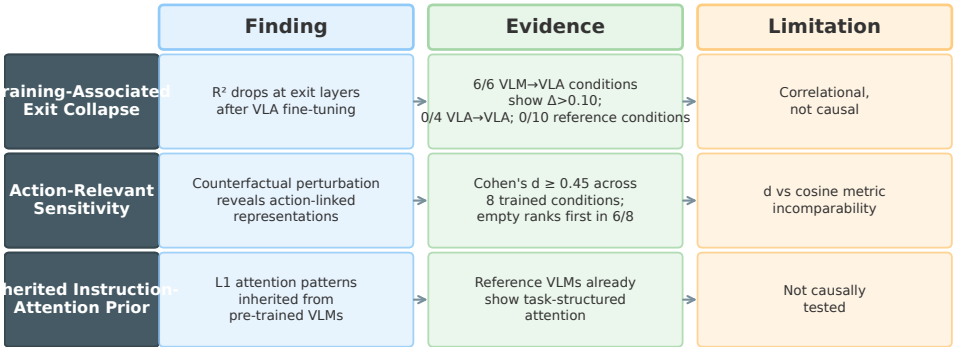



Figure 25: **Claim–evidence–scope summary.** Each main finding is paired with its supporting analysis and its main limitation. The figure distinguishes directly observed representational patterns from stronger causal interpretations that require interventional or training-dynamics evidence.

Despite these differences, both transitions exhibit the same exit-collapse pattern ($\Delta R^2 = 0.156$–$0.455$), while their respective reference VLMs show only small fluctuations ($\Delta R^2 \leq 0.097$). The probability of observing consistent collapse in two structurally divergent architectures under a null hypothesis of random association is low (permutation test $p = 0.008$; §A.19). Structural diversity thus provides evidential weight beyond what the raw $N = 2$ count suggests.

### A.39 Convergent Evidence Summary

Three methodologically distinct approaches triangulate instruction routing in VLAs. Table 33 summarizes what each method measures, its key finding, evidence type, and acknowledged limitation.

The three methods form a complementary chain: probing identifies *where* instruction information concentrates across layers, perturbation confirms *whether* instruction content functionally affects behavior, and ablation demonstrates *that* the identified attention structures carry relevant signals. No method alone supports causal claims about instruction routing mechanisms; their convergence provides stronger evidence than any single approach.

Table 29: Per-layer action prediction $R^2$: training effect across architectures (mean $\pm$ std over 15 seeds). Peak values per condition in **bold**.

| Layer | OpenVLA (Llama-2-7B) | | TraceVLA (Phi-3-mini) | |
|---|---|---|---|---|
| | **Reference** | **Trained** | **Reference** | **Trained** |
| L0 | $\mathbf{0.629 \pm 0.016}$ | $0.629 \pm 0.016$ | $-0.330 \pm 0.003$ | $-0.330 \pm 0.002$ |
| L1 | $0.539 \pm 0.023$ | $0.625 \pm 0.017$ | $0.429 \pm 0.034$ | $0.447 \pm 0.013$ |
| L2 | $0.517 \pm 0.047$ | $0.622 \pm 0.010$ | $0.458 \pm 0.023$ | $0.522 \pm 0.014$ |
| L3 | $0.448 \pm 0.216$ | $0.624 \pm 0.014$ | $0.498 \pm 0.036$ | $0.579 \pm 0.011$ |
| L4 | $0.496 \pm 0.057$ | $0.628 \pm 0.014$ | $0.572 \pm 0.021$ | $0.592 \pm 0.009$ |
| L5 | $0.475 \pm 0.101$ | $0.627 \pm 0.019$ | $0.573 \pm 0.020$ | $0.602 \pm 0.010$ |
| L6 | $0.422 \pm 0.119$ | $0.631 \pm 0.012$ | $\mathbf{0.578 \pm 0.018}$ | $0.609 \pm 0.012$ |
| L7 | $0.459 \pm 0.093$ | $0.632 \pm 0.012$ | $0.569 \pm 0.023$ | $0.616 \pm 0.014$ |
| L8 | $0.347 \pm 0.283$ | $0.635 \pm 0.013$ | $0.560 \pm 0.025$ | $0.615 \pm 0.010$ |
| L9 | $0.445 \pm 0.127$ | $\mathbf{0.637 \pm 0.008}$ | $0.564 \pm 0.021$ | $0.617 \pm 0.013$ |
| L10 | $0.353 \pm 0.224$ | $0.624 \pm 0.014$ | $0.551 \pm 0.027$ | $0.625 \pm 0.007$ |
| L11 | $0.356 \pm 0.122$ | $0.623 \pm 0.016$ | $0.561 \pm 0.025$ | $\mathbf{0.634 \pm 0.009}$ |
| L12 | $0.409 \pm 0.203$ | $0.626 \pm 0.011$ | $0.557 \pm 0.026$ | $0.632 \pm 0.005$ |
| L13 | $0.341 \pm 0.227$ | $0.624 \pm 0.013$ | $0.547 \pm 0.028$ | $0.633 \pm 0.008$ |
| L14 | $0.374 \pm 0.211$ | $0.623 \pm 0.015$ | $0.550 \pm 0.023$ | $0.631 \pm 0.009$ |
| L15 | $0.297 \pm 0.225$ | $0.619 \pm 0.012$ | $0.550 \pm 0.022$ | $0.633 \pm 0.010$ |
| L16 | $0.323 \pm 0.226$ | $0.619 \pm 0.011$ | $0.534 \pm 0.018$ | $0.631 \pm 0.010$ |
| L17 | $0.395 \pm 0.109$ | $0.609 \pm 0.023$ | $0.519 \pm 0.024$ | $0.633 \pm 0.007$ |
| L18 | $0.214 \pm 0.307$ | $0.613 \pm 0.017$ | $0.502 \pm 0.047$ | $0.629 \pm 0.012$ |
| L19 | $0.241 \pm 0.258$ | $0.613 \pm 0.013$ | $0.499 \pm 0.058$ | $0.627 \pm 0.008$ |
| L20 | $0.086 \pm 0.311$ | $0.611 \pm 0.014$ | $0.482 \pm 0.048$ | $0.628 \pm 0.011$ |
| L21 | $0.330 \pm 0.216$ | $0.589 \pm 0.023$ | $0.476 \pm 0.036$ | $0.624 \pm 0.007$ |
| L22 | $0.207 \pm 0.283$ | $0.586 \pm 0.013$ | $0.408 \pm 0.140$ | $0.616 \pm 0.008$ |
| L23 | $0.264 \pm 0.227$ | $0.578 \pm 0.028$ | $0.455 \pm 0.074$ | $0.613 \pm 0.007$ |
| L24 | $0.173 \pm 0.275$ | $0.577 \pm 0.013$ | $0.446 \pm 0.062$ | $0.610 \pm 0.008$ |
| L25 | $0.130 \pm 0.305$ | $0.553 \pm 0.053$ | $0.479 \pm 0.039$ | $0.609 \pm 0.010$ |
| L26 | $0.251 \pm 0.254$ | $0.535 \pm 0.046$ | $0.411 \pm 0.098$ | $0.611 \pm 0.008$ |
| L27 | $0.099 \pm 0.278$ | $0.552 \pm 0.027$ | $0.366 \pm 0.197$ | $0.604 \pm 0.011$ |
| L28 | $0.099 \pm 0.307$ | $0.544 \pm 0.034$ | $0.358 \pm 0.218$ | $0.596 \pm 0.015$ |
| L29 | $0.088 \pm 0.275$ | $0.530 \pm 0.034$ | $0.356 \pm 0.271$ | $0.585 \pm 0.010$ |
| L30 | $0.220 \pm 0.234$ | $0.511 \pm 0.029$ | $0.424 \pm 0.104$ | $0.574 \pm 0.008$ |
| L31 | $0.069 \pm 0.284$ | $0.488 \pm 0.054$ | $0.332 \pm 0.262$ | $0.557 \pm 0.016$ |
| L32 | $0.584 \pm 0.011$ | $0.535 \pm 0.015$ | $0.410 \pm 0.185$ | $0.550 \pm 0.007$ |

Table 30: **Methodological precedent.** Representative interpretability studies and their use of behavioral evaluation. Partial behavioral eval includes correlational or indirect evidence.

| Paper | Domain | Method | Behav. eval? |
|---|---|---|---|
| Belinkov [46] | NLP | Probing survey | No |
| Olsson et al. [23] | LM | Induction heads | No |
| Wang et al. [24] | LM | IOI circuit | No |
| Meng et al. [26] | LM | Causal tracing (localization) | No |
| Grant et al. [5] | VLA | Visual dominance | Partial |
| Haon et al. [6] | VLA | Sparse directions | No |
| Ours | VLA | $\Delta R^2$ + attn. profiling | Partial ($\rho$=0.583) |

Table 31: **NLP finetuning control comparison.** All three models are instruction-tuned for conversational use without action heads or robot demonstration data. None exhibits exit collapse.

| Model | VLA backbone | $\Delta R^2$ | Mean $R^2$ | Layers |
|---|---|---|---|---|
| Llama-2-7b-chat | OpenVLA (Llama-2-7B) | 0.000 | $0.155 \pm 0.001$ | 32 |
| Phi-3-mini-4k-instruct | TraceVLA (Phi-3-mini) | 0.000 | $0.155 \pm 0.000$ | 32 |
| Qwen2.5-3B-Instruct* | NORA (Qwen2.5-VL-3B) | 0.000 | $0.155 \pm 0.000$ | 36 |

*Qwen2.5-3B-Instruct shares the Qwen2.5 language backbone with NORA's Qwen2.5-VL-3B but lacks the vision encoder and cross-attention modules.

Table 32: **Structural diversity between the two VLM-to-VLA transitions.** The two architectures differ on nearly every major design axis, strengthening the generality of shared findings.

| Design Axis | TraceVLA | NORA |
|---|---|---|
| Language backbone | Phi-3-mini (3.8B) | Qwen2.5-VL (3B) |
| Backbone family | Microsoft | Alibaba |
| Attention mechanism | Multi-Head Attention | Grouped-Query Attention |
| Number of layers | 32 | 36 |
| Vision encoder | SigLIP (shared w/ base) | Native ViT (dynamic resolution) |
| Vision integration | Projector (late fusion) | Token concatenation (native) |
| Pretraining data | English-centric web | Multilingual web + code |
| Action head design | Linear projection | MLP |
| Training data | OXE + LIBERO | OXE + LIBERO + custom |

Table 33: **Convergent evidence across three analytical methods.** Each method addresses a different aspect of instruction routing; no single method is sufficient alone, but their convergence strengthens the overall characterization.

| Method | What it measures | Key finding | Evidence type | Limitation |
|---|---|---|---|---|
| Layer-wise probing (§4.3) | Where action-relevant information is decodable | VLM-to-VLA: peak-to-exit collapse ($\Delta R^2 = 0.16$–$0.46$); VLA-to-VLA: flat | Correlational (decodability) | Does not establish causal necessity |
| Counterfactual perturbation (§4.4) | Whether instruction content affects model outputs | All trained VLAs: graded divergence ($d = 0.45$–$1.52$) ordered by semantic distance | Behavioral (input–output) | Does not localize which layers are responsible |
| Attention head ablation (Appendix A.12) | Whether identified structures carry functional signals | Dose-response: monotonic divergence ($d = 0.48$–$0.64$) upon progressive head removal | Interventional (causal contribution) | Single architecture (OpenVLA); contribution $\neq$ necessity |